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(54) **OPTICAL LENS ASSEMBLY AND
HEAD-MOUNTED ELECTRONIC DEVICE**

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G02B 25/00 (2006.01)

G02B 27/01 (2006.01)

(52) **U.S. Cl.**

CPC **G02B 25/001** (2013.01); **G02B 27/0172**
(2013.01)

(58) **Field of Classification Search**

CPC G02B 25/001; G02B 27/0172

USPC 359/630, 716, 784

See application file for complete search history.

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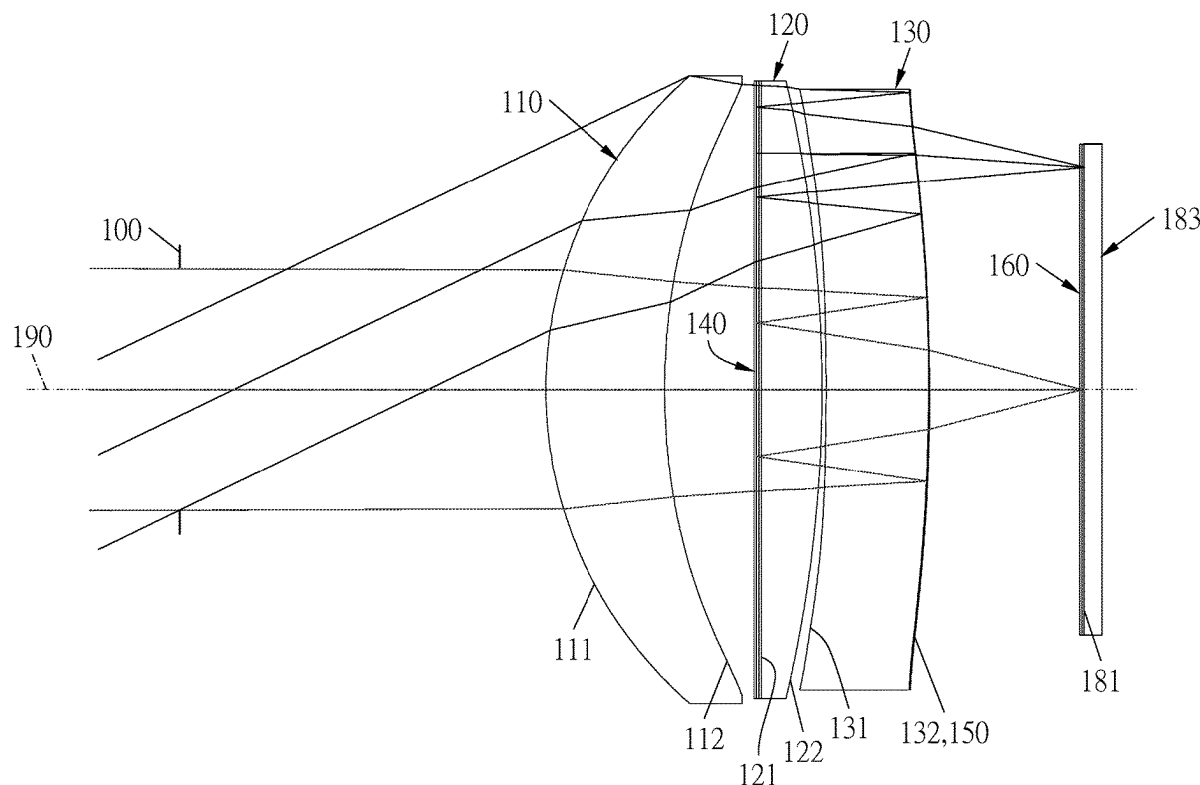
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(57) **ABSTRACT**

An optical lens assembly includes, an optical element group;
and in order from a visual side to an image source side: a first
lens with positive refractive power, a second lens with
refractive power, a third lens with refractive power, and an
image source-side surface of the third lens being convex in
a paraxial region thereof, and a partial-reflective-partial-
transmissive element. Wherein the optical element group is
disposed between the first lens and the third lens, and
includes, in order from the visual side to the image source
side: an absorptive polarizer, a reflective polarizer and a
phase retarder. The optical lens assembly may become
lightweight and have good image quality when satisfying a
specific condition.

20 Claims, 11 Drawing Sheets



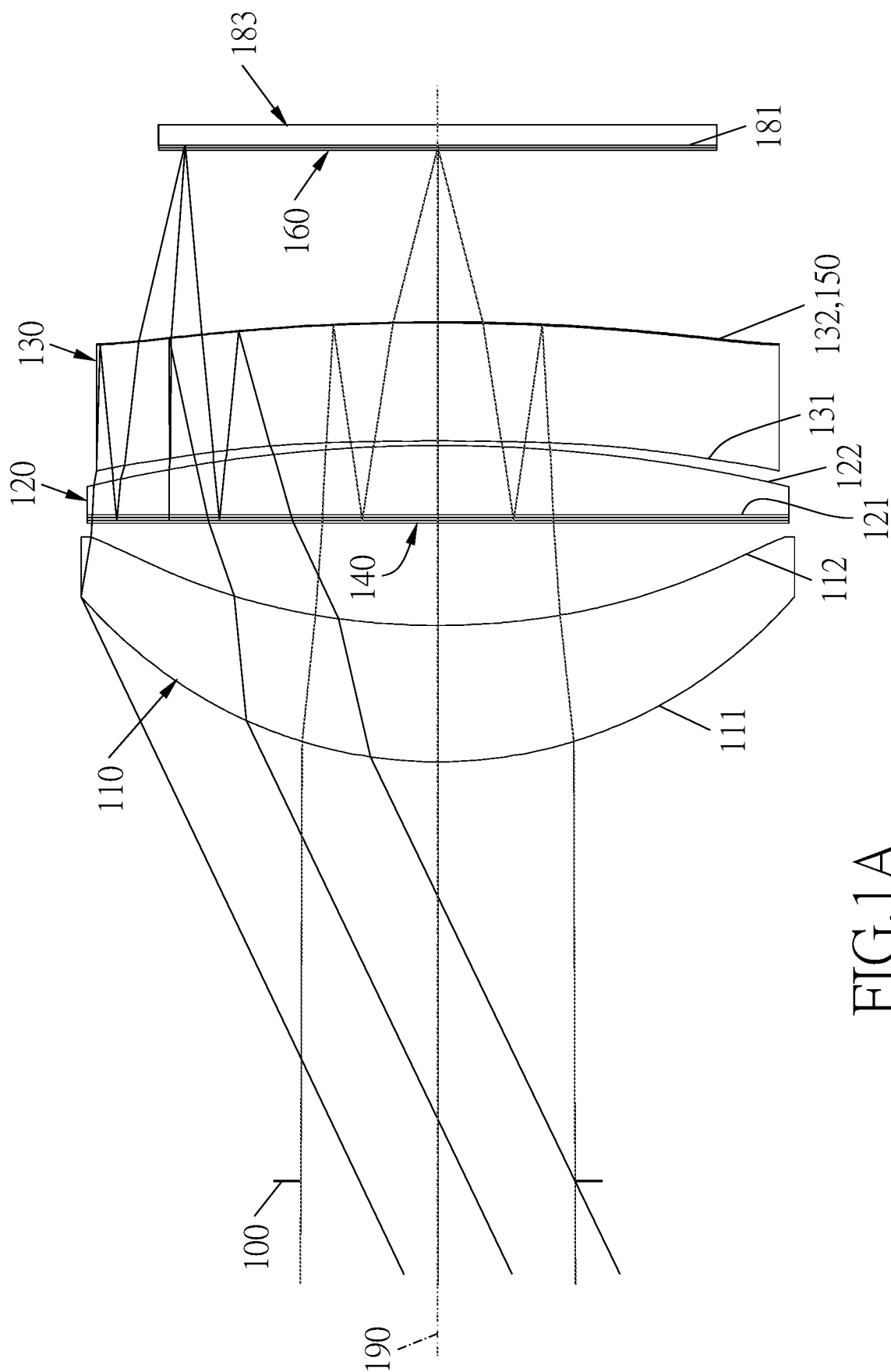


FIG.1A

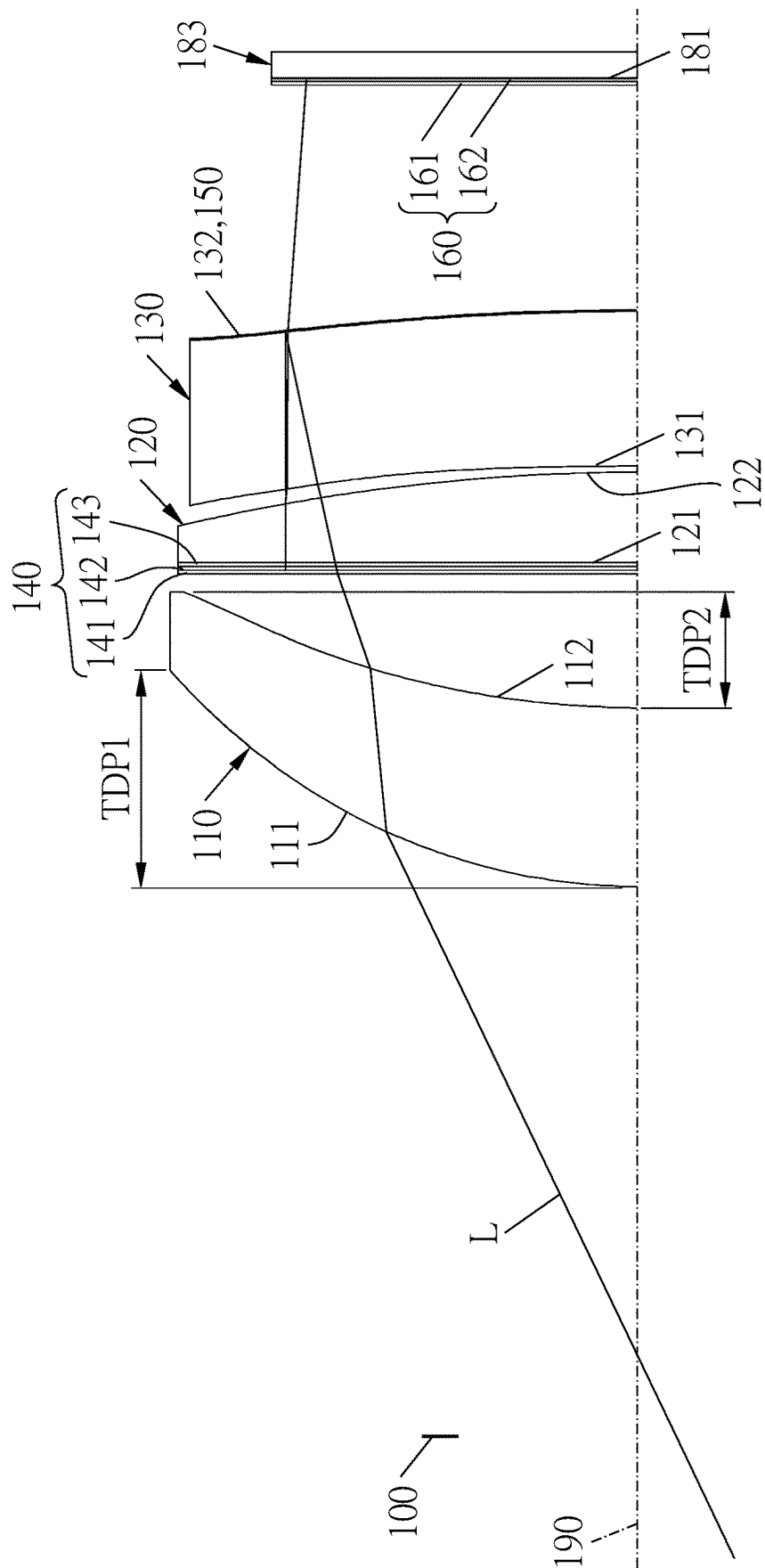


FIG. 1B

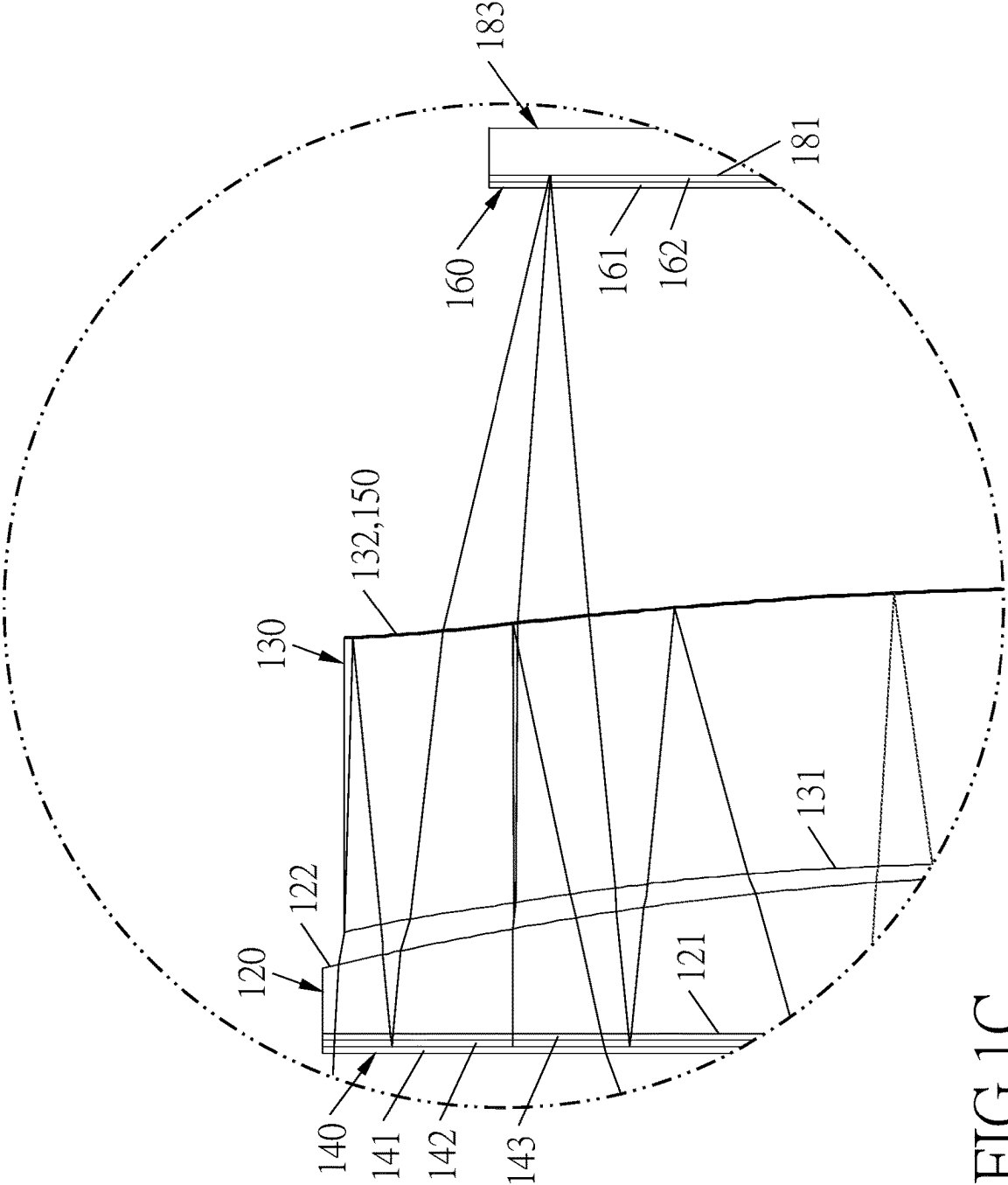


FIG. 1C

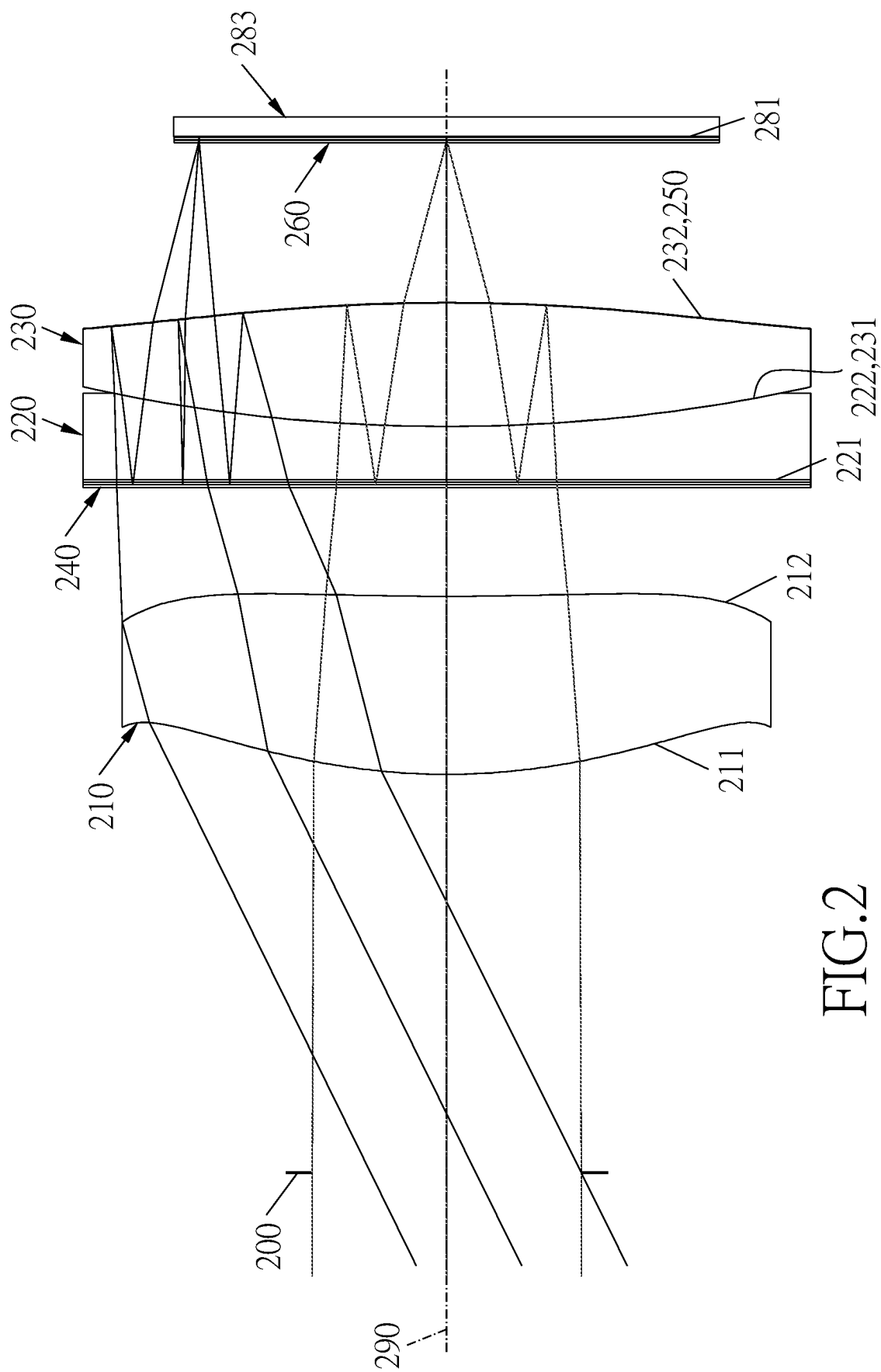


FIG. 2

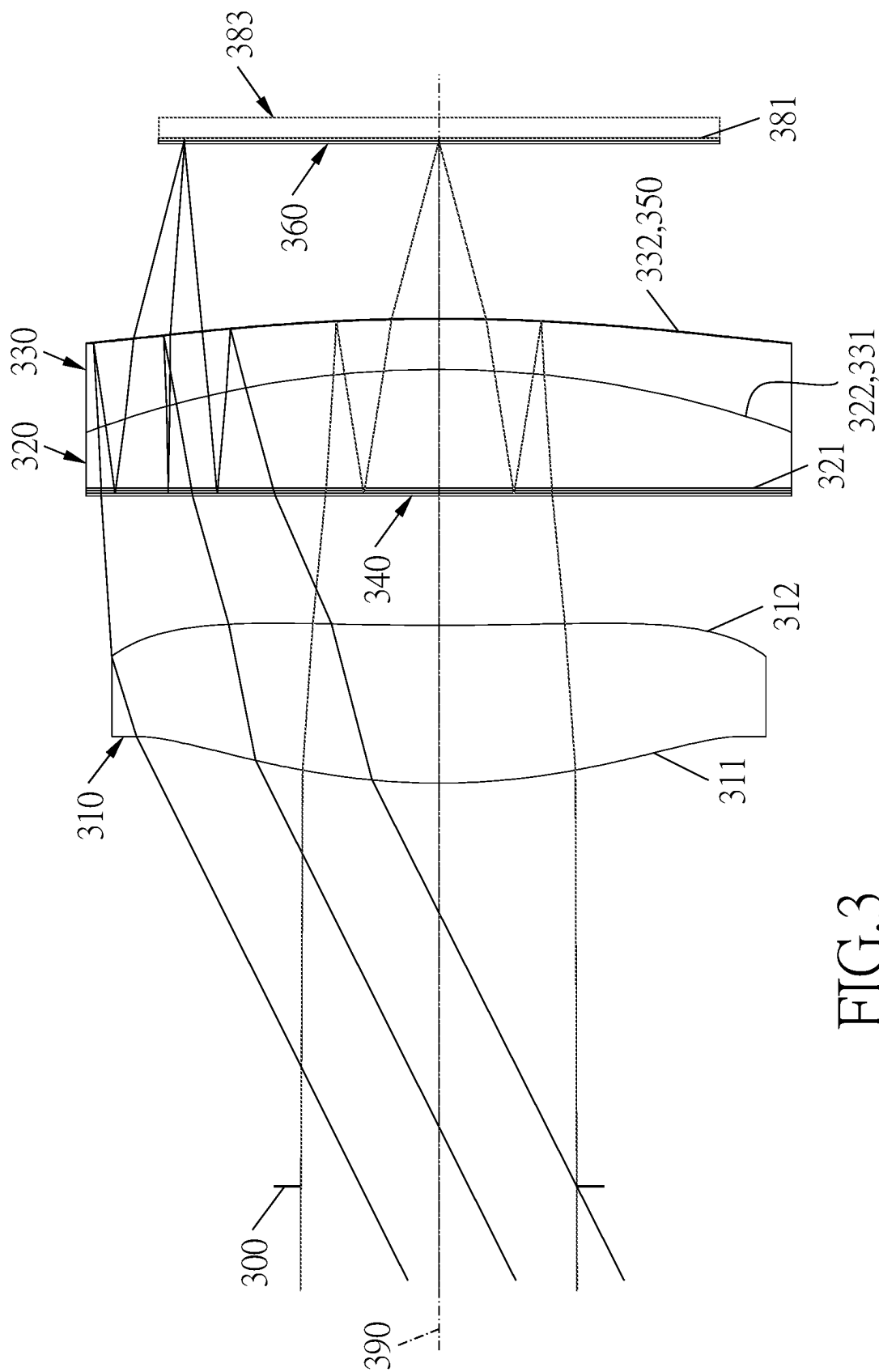


FIG.3

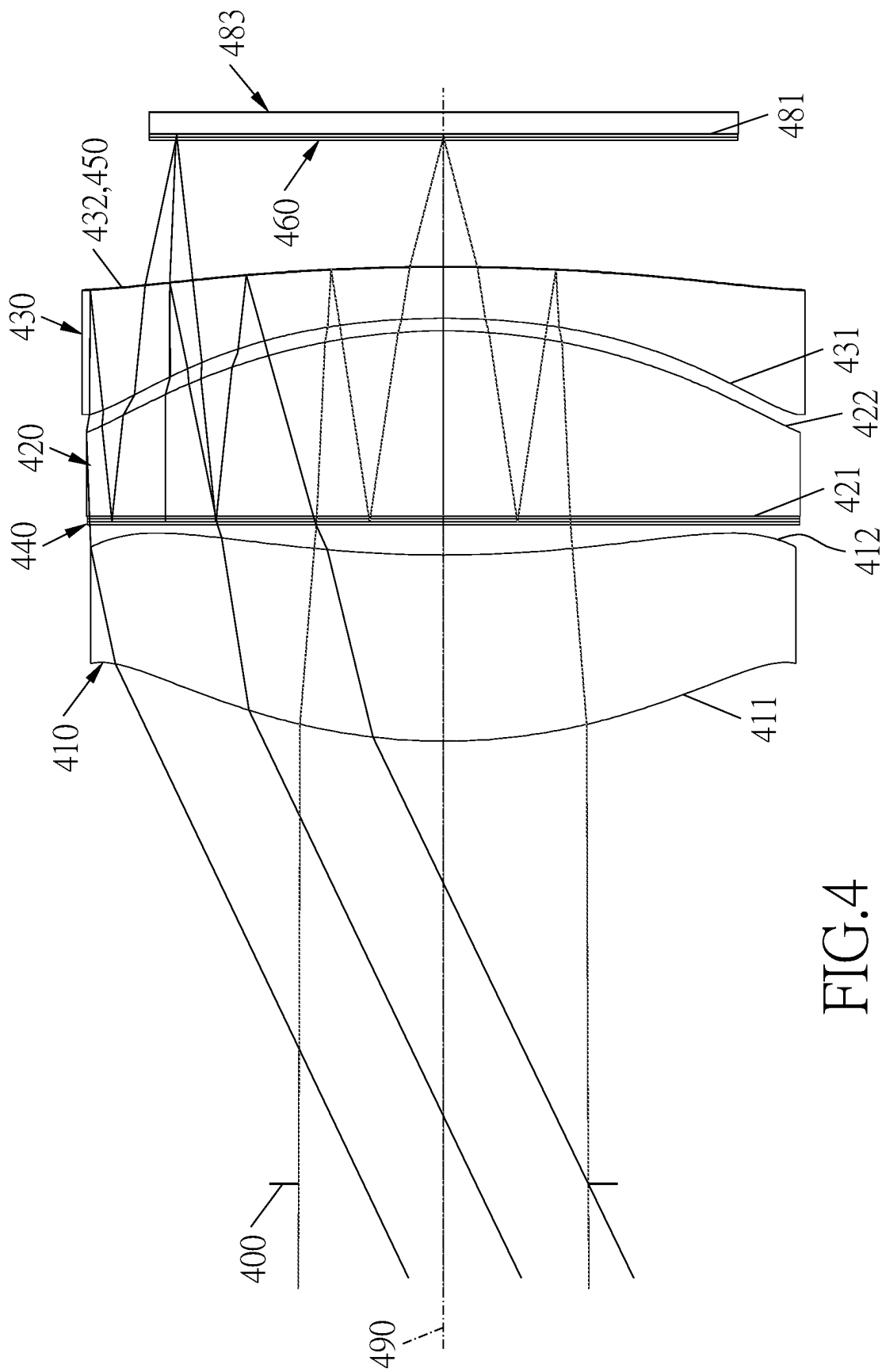


FIG.4

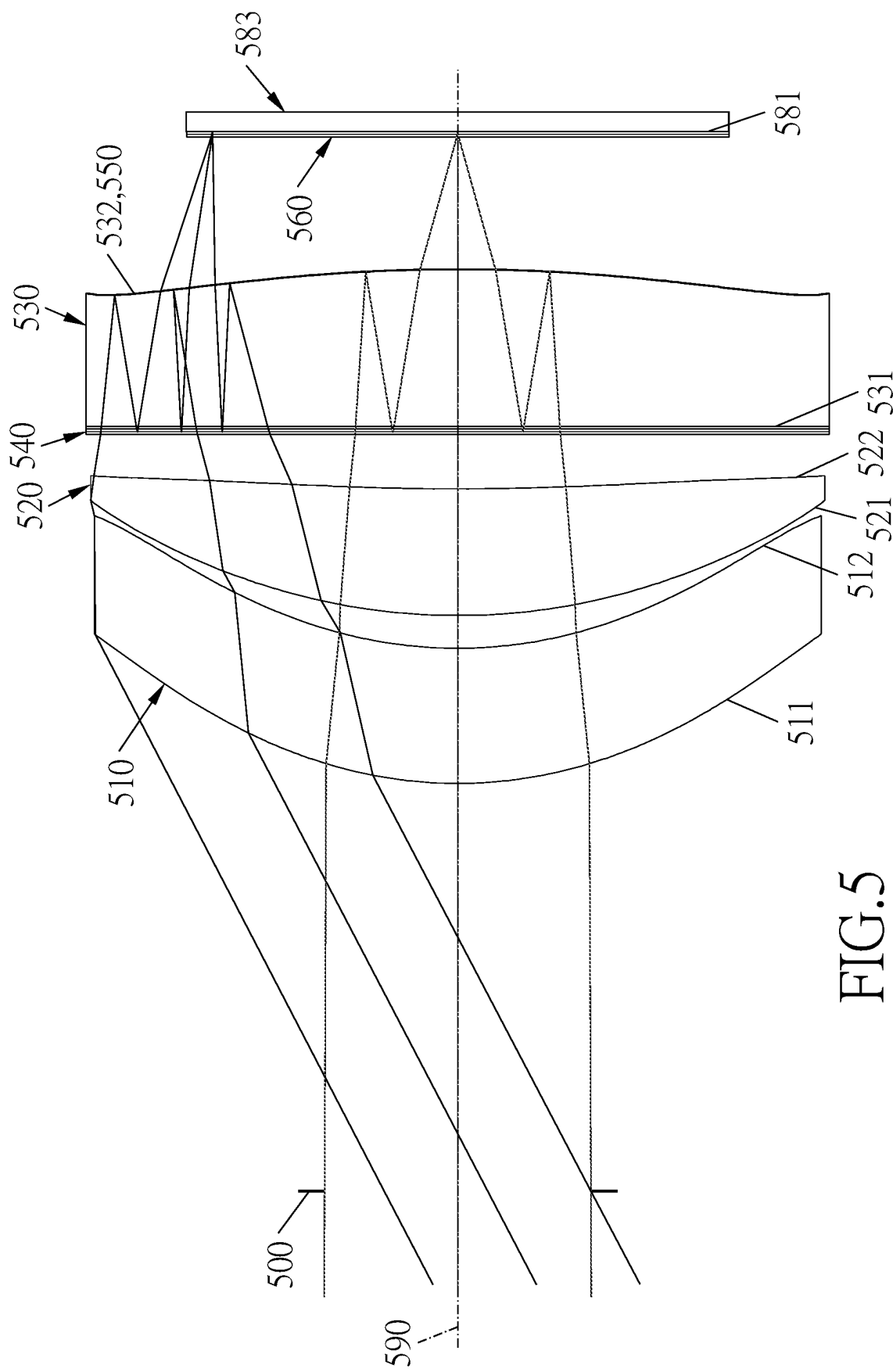


FIG. 5

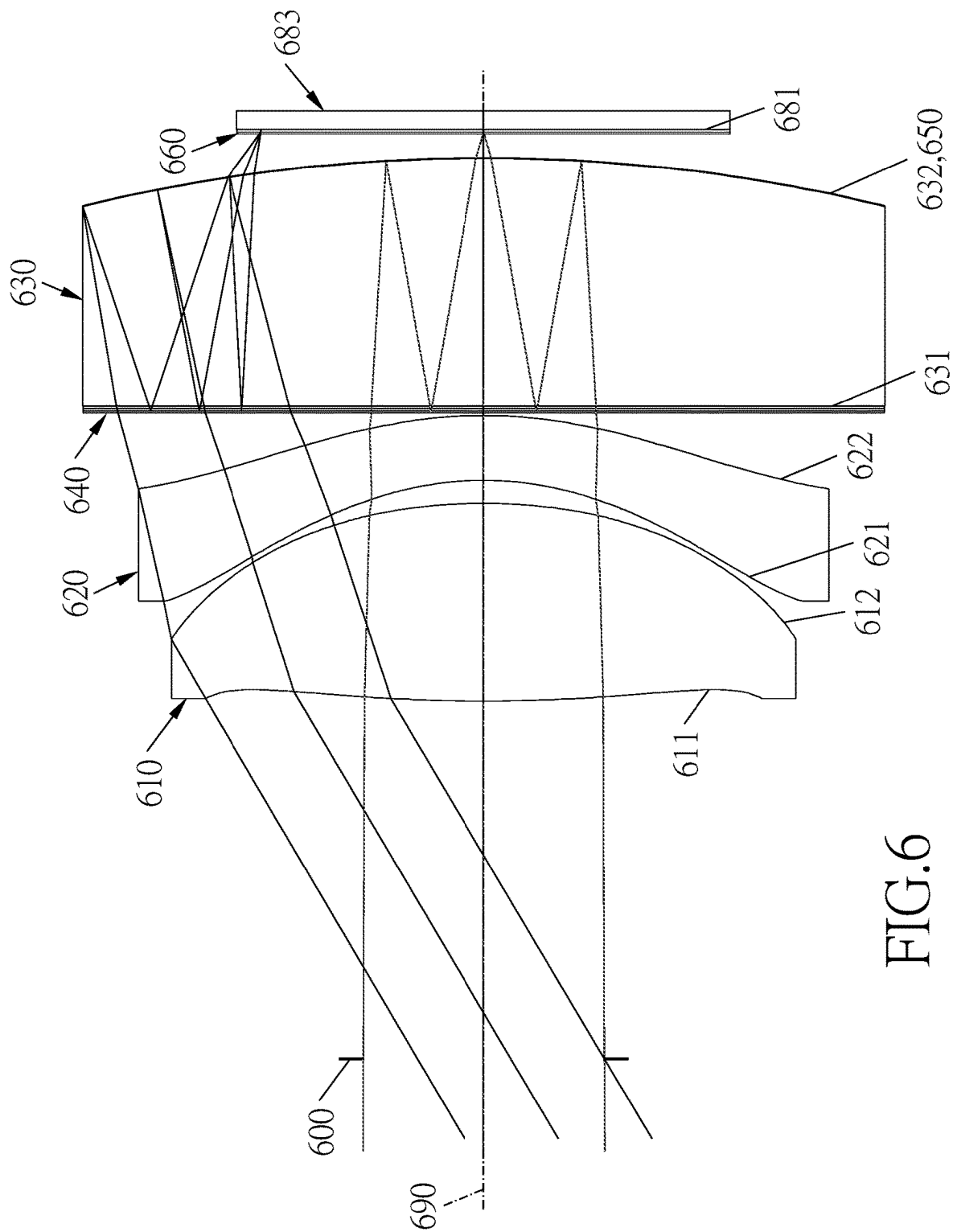


FIG. 6

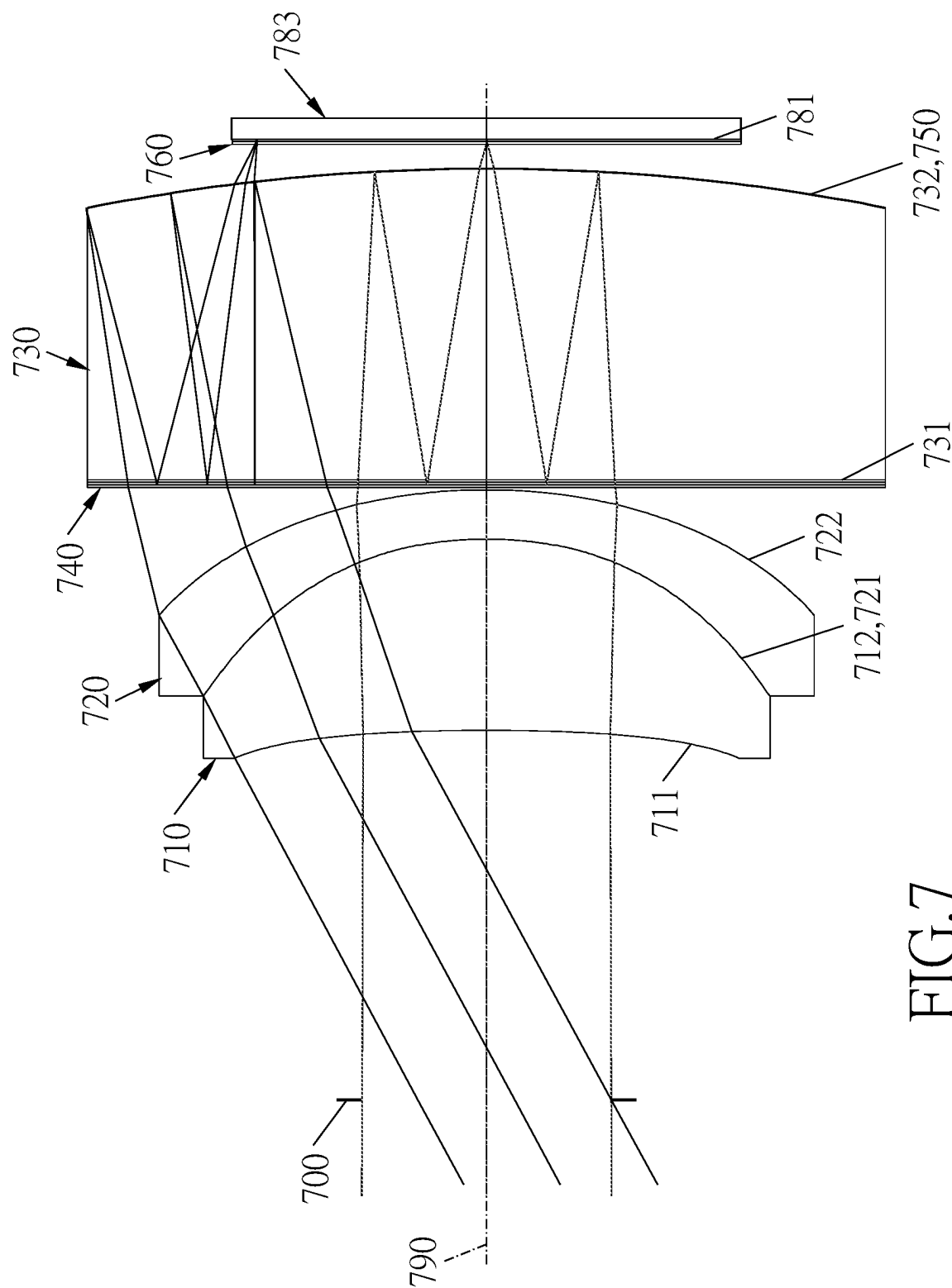


FIG. 7

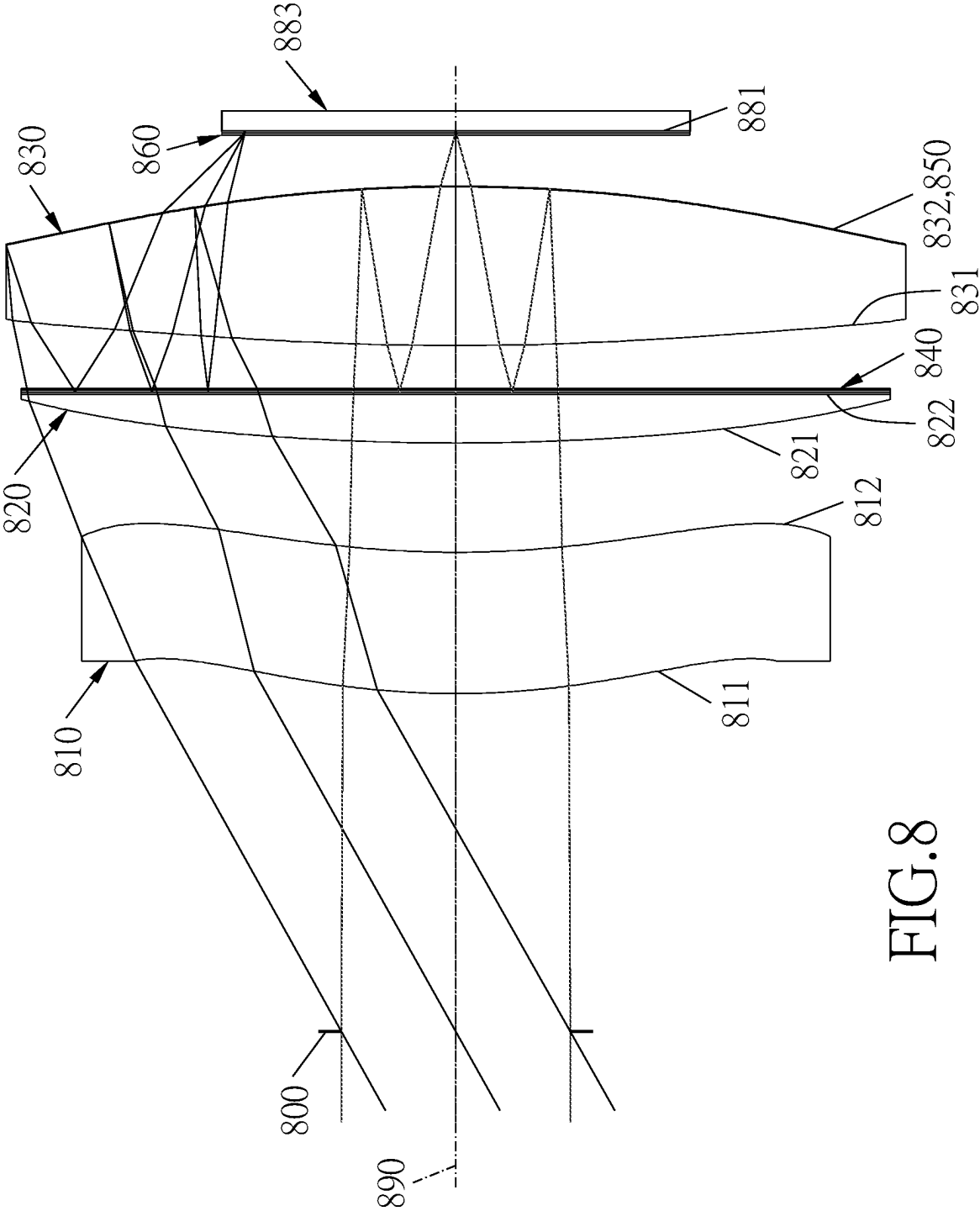


FIG. 8

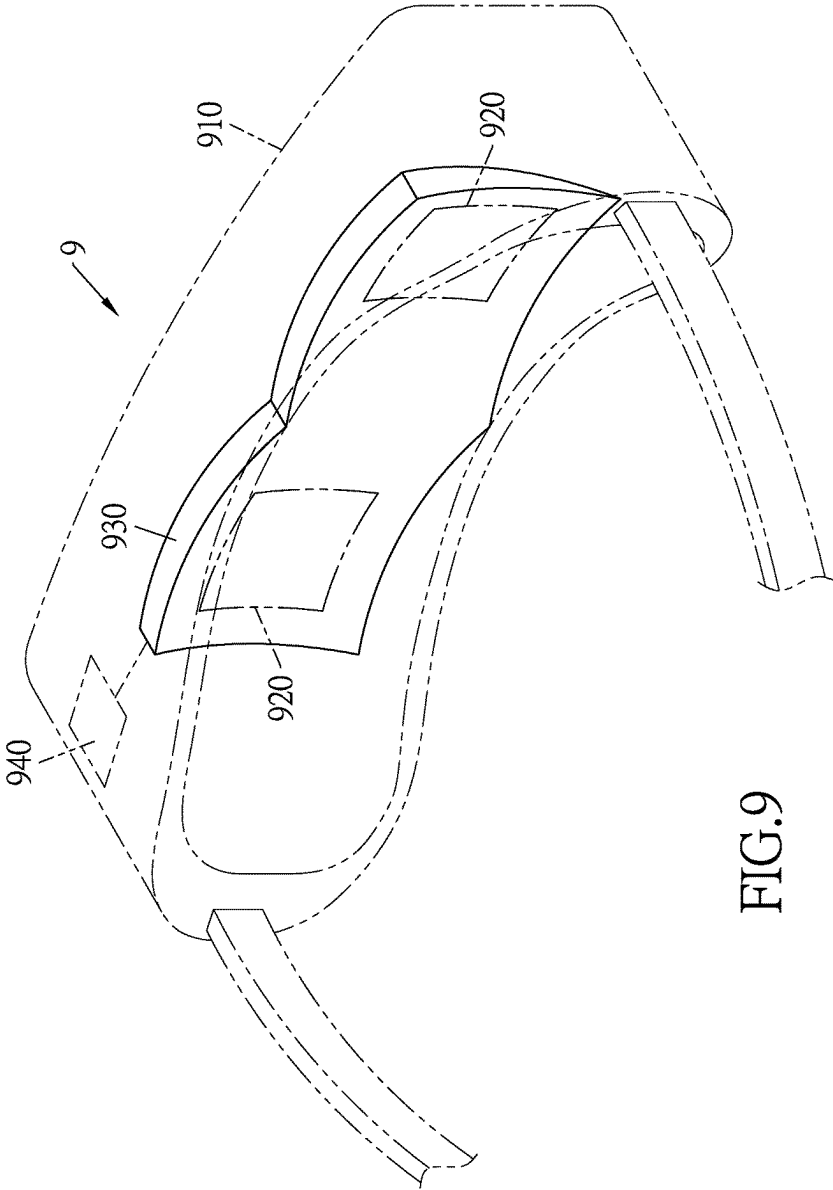


FIG.9

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OPTICAL LENS ASSEMBLY AND HEAD-MOUNTED ELECTRONIC DEVICE

BACKGROUND

Field of the Invention

The present invention relates to an optical lens assembly and head-mounted electronic device, and more particularly to an optical lens assembly applicable to head-mounted electronic devices.

Description of Related Art

With the development of the semiconductor industry, the functions of various consumer electronic products are increasingly powerful. Moreover, various services of the software application end emerge. These enable consumers to have more choices. Virtual reality (VR) technology, Augmented reality (AR) technology and Mixed reality (MR) technology emerge when the market is no longer satisfied with handheld electronic products. Nowadays, the application of virtual reality opens up a blue ocean market for consumer electronics, and in the application field of virtual reality, the first commercialized project is the head-mounted display.

However, the current head-mounted displays are heavy and have poor image quality.

The present invention mitigates and/or obviates the aforementioned disadvantages.

SUMMARY

The objective of the present invention is to provide an optical lens assembly and a head-mounted electronic device, which can reduce the number of lenses by folding the light path, so as to reduce the weight of the device, and also can provide better image quality.

Therefore, an optical lens assembly in accordance with an embodiment of the present invention includes: an optical element group; and in order from a visual side to an image source side: a first lens with positive refractive power, a second lens with refractive power, a third lens with refractive power, and an image source-side surface of the third lens being convex in a paraxial region thereof, and a partial-reflective-partial-transmissive element. Wherein the optical element group is disposed between the first lens and the third lens, and includes, in order from the visual side to the image source side: an absorptive polarizer (that is, a first absorptive polarizer), a reflective polarizer and a phase retarder (that is, a first phase retarder). A radius of curvature of an image source-side surface of the first lens is R_2 , a thickness of the first lens along an optical axis is CT_1 , a focal length of the first lens is f_1 , a focal length of the optical lens assembly is f , and the following condition is satisfied: $-5.88 < (R_2/CT_1) * (f_1/f) < 110.22$.

When $-5.88 < (R_2/CT_1) * (f_1/f) < 110.22$ is satisfied, the distortion of the optical lens assembly can be effectively improved, the aberration of the optical lens assembly can be reduced, and the size of the lens can be further reduced.

Optionally, a focal length of the third lens is f_3 , the focal length of the optical lens assembly is f , and the following condition is satisfied: $-6.92 < f_3/f < 23.80$, so that the ratio of the focal length of the third lens to that of the optical lens assembly can enhance the wide-field of view characteristic

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of the optical lens assembly, thereby providing a larger field of view and maintaining the illumination of the optical lens assembly.

Optionally, the focal length of the third lens is f_3 , a radius of curvature of a visual-side surface of the first lens is R_1 , and the following condition is satisfied: $-5.58 < f_3/R_1 < 27.18$, which can effectively improve the aberration of the optical lens assembly, further improving the image quality.

Optionally, a maximum effective radius of the visual-side surface of the first lens is CA_1 , an absolute value of a displacement in parallel to the optical axis from an intersection between the visual-side surface of the first lens and the optical axis to the maximum effective radius position on the visual-side surface of the first lens is TDP_1 , and the following condition is satisfied: $1.76 < CA_1/TDP_1 < 137.17$, which can achieve a larger field of view and reduce the size of the lens.

Optionally, a maximum effective radius of the image source-side surface of the first lens is CA_2 , an absolute value of a displacement in parallel to the optical axis from an intersection between the image source-side surface of the first lens and the optical axis to the maximum effective radius position on the image source-side surface of the first lens is TDP_2 , and the following condition is satisfied: $1.43 < CA_2/TDP_2 < 56.16$, which can achieve a larger field of view and reduce the size of the lens.

Optionally, a radius of curvature of the image source-side surface of the third lens is R_6 , a thickness of the third lens along the optical axis is CT_3 , and the following condition is satisfied: $-58.44 < R_6/CT_3 < -5.15$, which is conducive to achieving a proper balance between the radius of curvature and the thickness of the third lens.

Optionally, the radius of curvature of the visual-side surface of the first lens is R_1 , the radius of curvature of the image source-side surface of the first lens is R_2 , the focal length of the first lens is f_1 , and the following condition is satisfied: $-32.71 \text{ mm} < (R_1 * R_2)/f_1 < 60.84 \text{ mm}$, which is conducive to achieving the more appropriate distribution of the radius of curvature of the first lens, thereby reducing the chromatic aberration.

Optionally, the radius of curvature of the visual-side surface of the first lens is R_1 , the radius of curvature of the image source-side surface of the first lens is R_2 , and the following condition is satisfied: $-3.55 < R_1/R_2 < 6.52$, which is conducive to preventing the radius of curvature from being too small and reducing the sensitivity to the assembly tolerance as the two radii of curvature of the first lens are conditioned by each other.

Optionally, the radius of curvature of the image source-side surface of the third lens is R_6 , the radius of curvature of the image source-side surface of the first lens is R_2 , and the following condition is satisfied: $-4.59 < R_6/R_2 < 6.08$, which is conducive to preventing the radius of curvature from being too small and reducing the sensitivity to the assembly tolerance as the two radii of curvature are conditioned by each other.

Optionally, a thickness of the second lens along the optical axis is CT_2 , the thickness of the third lens along the optical axis is CT_3 , and the following condition is satisfied: $0.22 < CT_3/CT_2 < 7.58$, which is conducive to ensuring that the thickness of the optical lens assembly can meet the processing requirement of the manufacturing process of the lens device, while ensuring the image quality.

Optionally, the thickness of the first lens along the optical axis is CT_1 , the thickness of the second lens along the optical axis is CT_2 , the thickness of the third lens along the optical axis is CT_3 , and the following condition is satisfied:

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$0.62 < (CT1 + CT2) / CT3 < 8.67$, which is conducive to improving the thickness matching between the lenses and ensuring the assembly stability, while ensuring the image quality.

Optionally, the radius of curvature of the visual-side surface of the first lens is $R1$, the thickness of the first lens along the optical axis is $CT1$, and the following condition is satisfied: $-13.48 < R1 / CT1 < 9.76$, so that the distortion of the optical lens assembly can be effectively improved, the aberration of the optical lens assembly can be reduced, and the size of the lens can be further reduced.

Optionally, the focal length of the first lens is $f1$, the focal length of the optical lens assembly is f , and the following condition is satisfied: $1.42 < f1 / f < 16.1$, so that the ratio of the focal length of the first lens to that of the optical lens assembly can enhance the wide-field of view characteristic of the optical lens assembly, thereby providing a larger field of view and maintaining the illumination of the optical lens assembly.

Optionally, the maximum effective radius of the image source-side surface of the first lens is $CA2$, a distance from the visual-side surface of the first lens to an image source plane along the optical axis is TL , the radius of curvature of the visual-side surface of the first lens is $R1$, and the following condition is satisfied: $-49.53 \text{ mm} < (CA2 / TL) * R1 < 43.8 \text{ mm}$, which is conducive to achieving a proper balance between the radius of curvature, the size of the first lens, and the length of the optical lens assembly.

Optionally, the focal length of the first lens is $f1$, the focal length of the third lens is $f3$, and the following condition is satisfied: $-1.10 < f1 / f3 < 3.53$, which is conducive to achieving the more appropriate distribution of the refractive power of the optical lens assembly, thereby reducing the aberration.

Optionally, a focal length of the second lens is $f2$, the focal length of the optical lens assembly is f , and the following condition is satisfied: $-5.80 < f2 / f < 13.35$, so that the ratio of the focal length of the second lens to that of the optical lens assembly can enhance the wide-field of view characteristic of the optical lens assembly, thereby providing a larger field of view and maintaining the illumination of the optical lens assembly.

Optionally, the radius of curvature of the image source-side surface of the third lens is $R6$, a radius of curvature of a visual-side surface of the third lens is $R5$, and the following condition is satisfied: $-1.37 < R6 / R5 < 3.67$, which is conducive to preventing the radius of curvature from being too small and reducing the sensitivity to the assembly tolerance as the two radii of curvature of the third lens are conditioned by each other.

Moreover, a head-mounted electronic device in accordance with an embodiment of the present invention includes a housing, the aforementioned optical lens assembly disposed in the housing, an image source disposed on the image source plane of the optical lens assembly in the housing, and a controller disposed in the housing and electrically connected to the image source.

The present invention will be presented in further details from the following descriptions with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiments in accordance with the present invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a schematic view of an optical lens assembly in accordance with a first embodiment of the present invention;

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FIG. 1B is a schematic diagram of a part of the optical lens assembly of FIG. 1A with a light path of a chief ray therein;

FIG. 1C is a partial amplified view of FIG. 1A;

FIG. 2 is a schematic view of an optical lens assembly in accordance with a second embodiment of the present invention;

FIG. 3 is a schematic view of an optical lens assembly in accordance with a third embodiment of the present invention;

FIG. 4 is a schematic view of an optical lens assembly in accordance with a fourth embodiment of the present invention;

FIG. 5 is a schematic view of an optical lens assembly in accordance with a fifth embodiment of the present invention;

FIG. 6 is a schematic view of an optical lens assembly in accordance with a sixth embodiment of the present invention;

FIG. 7 is a schematic view of an optical lens assembly in accordance with a seventh embodiment of the present invention;

FIG. 8 is a schematic view of an optical lens assembly in accordance with an eighth embodiment of the present invention; and

FIG. 9 is a schematic diagram of a head-mounted electronic device in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

First Embodiment

Referring to FIGS. 1A, 1B and 1C, an optical lens assembly in accordance with a first embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis 190: a stop 100, a first lens 110, a first optical element group 140, a second lens 120, a third lens 130, a partial-reflective-partial-transmissive element 150, a second optical element group 160 and an image source plane 181. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop 100 may be located in a position where the user's eyes view an image.

The first lens 110 with positive refractive power includes a visual-side surface 111 and an image source-side surface 112, the visual-side surface 111 of the first lens 110 is convex in a paraxial region thereof, the image source-side surface 112 of the first lens 110 is concave in a paraxial region thereof, the visual-side surface 111 and the image source-side surface 112 of the first lens 110 are aspheric, and the first lens 110 is made of plastic.

The second lens 120 with positive refractive power includes a visual-side surface 121 and an image source-side surface 122, the visual-side surface 121 of the second lens 120 is flat in a paraxial region thereof, the image source-side surface 122 of the second lens 120 is convex in a paraxial region thereof, the image source-side surface 122 of the second lens 120 is spherical, and the second lens 120 is made of plastic.

The third lens 130 with positive refractive power includes a visual-side surface 131 and an image source-side surface 132, the visual-side surface 131 of the third lens 130 is concave in a paraxial region thereof, the image source-side surface 132 of the third lens 130 is convex in a paraxial region thereof, the visual-side surface 131 and the image

source-side surface **132** of the third lens **130** are aspheric, and the third lens **130** is made of plastic.

The first optical element group **140** is disposed on the visual-side surface **121** of the second lens **120** and includes, in order from the visual side to the image source side along the optical axis **190**: a first absorptive polarizer **141**, a reflective polarizer **142** and a first phase retarder **143**. The first phase retarder **143** is disposed on the visual-side surface **121** of the second lens **120** and is, for example, but not limited to, a quarter-wave plate. The reflective polarizer **142** is disposed on the first phase retarder **143**. The first absorptive polarizer **141** is disposed on the reflective polarizer **142**. The first optical element group **140** is located between the first lens **110** and the third lens **130**.

The partial-reflective-partial-transmissive element **150** is disposed on the image source-side surface **132** of the third lens **130** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **150** for different wavelengths.

The second optical element group **160** is disposed on the image source plane **181** and includes, in order from the visual side to the image source side along the optical axis **190**: a second phase retarder **161** and a second absorptive polarizer **162**. The second absorptive polarizer **162** is disposed on the image source plane **181**. The second phase retarder **161** is disposed on the second absorptive polarizer **162** and is, for example, but not limited to, a quarter-wave plate.

The optical lens assembly works in cooperation with an image source **183** disposed on the image source plane **181**. In the present embodiment, the type of the image source **183** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

The curve equation for the aspheric surface profiles of the respective lenses of the first embodiment is expressed as follows:

$$z(h) = \frac{ch^2}{1 + [1 - (k+1)c^2h^2]^{0.5}} + \sum (A_i) \cdot (h^i)$$

wherein:

z represents the value of a reference position at a height of h with respect to a vertex of the surface of a lens along the optical axis **190**;

c represents a paraxial curvature (i.e., a curvature of a lens surface in a paraxial region thereof) equal to 1/R (R: a paraxial radius of curvature);

h represents a vertical distance from the point on the curve of the aspheric surface to the optical axis **190**;

k represents the conic constant; and

A_i represents the i-th order aspheric coefficient.

The optical lens assembly of the first embodiment utilizes the configuration and arrangement of the absorptive polarizer, the reflective polarizer, the phase retarders, the partial-reflective-partial-transmissive element and the lenses to fold the light path thereof by the transmission and reflection of light to shorten the length of the optical lens assembly required for forming an image without affecting the image quality. In a light path L in FIG. 1B, an unpolarized beam from the image source **183** turns to a linearly-polarized beam after passing through the second absorptive polarizer **162**, the linearly-polarized beam turns to a circularly-polarized beam after passing through the second phase retarder

161, and the circularly-polarized beam then transmits to the partial-reflective-partial-transmissive element **150**. A component of the circularly-polarized beam passes through the partial-reflective-partial-transmissive element **150**, and then passes through the third lens **130**, the second lens **120** and the first phase retarder **143** sequentially to turn to a linearly-polarized beam with a polarization direction parallel to the reflective axis of the reflective polarizer **142** and transmit to the reflective polarizer **142**. This linearly-polarized beam is reflected by the reflective polarizer **142** and passes through the first phase retarder **143** again to turn to a circularly-polarized beam, and then passes through the second lens **120** and the third lens **130** sequentially to transmit to partial-reflective-partial-transmissive element **150**. Then, a component of the circularly-polarized beam is reflected by the partial-reflective-partial-transmissive element **150**, and then passes through the third lens **130**, the second lens **120** and the first phase retarder **143** sequentially to turn to a linearly-polarized beam with a polarization direction vertical to the reflective axis of the reflective polarizer **142**. Finally, the linearly-polarized beam transmits to the user's eyes after passing through the reflective polarizer **142**, the first absorptive polarizer **141** and the first lens **110** sequentially.

Please refer to Tables 1-4, Table 1 shows the detailed optical data of the elements of the optical lens assembly of the first embodiment, Table 2 shows the aspheric coefficients of the aspherical surfaces of the elements of the optical lens assembly of the first embodiment, Table 3 shows the remaining parameters of the optical lens assembly of the first embodiment and the values thereof, and the values of the parameters in Tables 1 and 3 meet the conditional formulas of Table 4. A focal length of the first lens **110** is f₁, a focal length of the second lens **120** is f₂, a focal length of the third lens **130** is f₃, a thickness of the first lens **110** along the optical axis **190** is CT1, a thickness of the second lens **120** along the optical axis **190** is CT2, a thickness of the third lens **130** along the optical axis **190** is CT3, a maximum effective radius of the visual-side surface **111** of the first lens **110** is CA1, a maximum effective radius of the image source-side surface **112** of the first lens **110** is CA2, an absolute value of a displacement in parallel to the optical axis **190** from an intersection between the visual-side surface **111** of the first lens **110** and the optical axis to the maximum effective radius position on the visual-side surface **111** of the first lens **110** is TDP1, an absolute value of a displacement in parallel to the optical axis **190** from an intersection between the image source-side surface **112** of the first lens **110** and the optical axis **190** to the maximum effective radius position on the image source-side surface **112** of the first lens **110** is TDP2, a radius of curvature of the visual-side surface **111** of the first lens **110** is R1, a radius of curvature of the image-side surface **112** of the first lens **110** is R2, a radius of curvature of the visual-side surface of **131** the third lens **130** is R5, a radius of curvature of the image source-side surface **132** of the third lens **130** is R6, a distance from the visual-side surface **111** of the first lens **110** to the image source plane **181** along the optical axis **190** is TL.

TABLE 1

Embodiment 1						
f = 19.65 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.08°						
Surface		Radius of curvature	Thickness/gap	Refractive index (nd)	Abbe number (vd)	Refraction/reflection
0	Stop	Infinity	15.000	—	—	—
1	First lens	17.211	4.879	1.544	55.9	Refraction
2		27.594	3.660	—	—	Refraction
3	First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
4	Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction
5	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
6	Second lens	Infinity	2.475	1.544	55.9	Refraction
7		-56.364	0.165	—	—	Refraction
8	Third lens	-146.533	4.238	1.544	55.900	Refraction
9	Partial-reflective-partial-transmissive element	-87.756	-4.238	—	—	Reflection
10		-56.364	-0.165	—	—	Refraction
11	Second lens	-146.533	-2.475	1.544	55.900	Refraction
12	First phase retarder	Infinity	-0.100	1.533	56.0	Refraction
13	Reflective polarizer	Infinity	-0.100	1.533	56.0	Refraction
14	Reflective polarizer	Infinity	0.100	—	—	Reflection
15	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
16	Second lens	Infinity	2.475	1.544	55.900	Refraction
17		-56.364	0.165	—	—	Refraction
18	Third lens	-146.533	4.238	1.544	55.9	Refraction
19	Partial-reflective-partial-transmissive element	-87.756	6.152	—	—	Refraction
20	Second phase retarder	Infinity	0.100	1.533	56.0	Refraction
21	Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
22	Image source plane	Infinity	—	—	—	—

The reference wavelength is 550 nm.

TABLE 2

Embodiment 1						
Aspheric Coefficients						
Surface	1	2	6, 12, 16	7, 10, 17	8, 11, 18	9, 19
K:	-1.2068E-02	2.3959E-01	0.0000E+00	0.0000E+00	9.0000E+01	0.0000E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	-2.2912E-06	7.2200E-06	0.0000E+00	0.0000E+00	-3.7084E-05	-3.7367E-06
A6:	4.2764E-08	4.5178E-08	0.0000E+00	0.0000E+00	1.3923E-08	-1.2232E-09
A8:	-7.8998E-11	-1.0456E-10	0.0000E+00	0.0000E+00	4.0522E-10	2.1979E-10
A10:	-7.8962E-13	-4.1630E-13	0.0000E+00	0.0000E+00	4.2746E-12	5.3435E-13
A12:	-9.0492E-16	-4.2221E-15	0.0000E+00	0.0000E+00	1.8432E-14	-2.2481E-15
A14:	8.7059E-18	-4.0336E-17	0.0000E+00	0.0000E+00	-2.2142E-17	2.5722E-17
A16:	5.5507E-21	-8.6824E-20	0.0000E+00	0.0000E+00	-1.1992E-18	0.0000E+00
A18:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 3

Embodiment 1					
f1 [mm]	71.67	CA1 [mm]	13.01	TDP1 [mm]	5.91
f2 [mm]	102.99	CA2 [mm]	12.64	TDP2 [mm]	3.16
f3 [mm]	389.82	—	—	—	—

TABLE 4

Embodiment 1			
f1/f	3.65	f3/f	19.83
R1/R2	0.62	R6/R5	0.60
R6/R2	-3.18	R6/CT3	-20.71
CT3/CT2	1.71	(R1*R2)/f1	1.22
(CT1 + CT2)/CT3	1.74	CA1/TDP1	2.20
R1/CT1	3.53	CA2/TDP2	4.01

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TABLE 4-continued

Embodiment 1			
f3/R1	22.65	(CA2/TL)*R1	9.86
f1/f3	0.18	(R2/CT1)(f1/f)	20.62
f2/f	5.24	—	—

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In Table 1, the units of the radius of curvature, the thickness, the gap and the focal length are expressed in mm, and the surface numbers 22-0 respectively represent the surfaces to which the light sequentially transmits from the image source plane **181** to the stop **100** along the light path L, wherein the surface 0 represents a gap between the stop **100** (or the user's eyes) and the visual-side surface **111** of the first lens **110** along the optical axis **190**; the surface 1 represents the thickness of the first lens **110** along the optical axis **190**; the surface 2 represents a gap between the image source-side surface **112** of the first lens **110** and the first

absorptive polarizer **141** along the optical axis **190**; the surface 3 represents the thickness of the first absorptive polarizer **141** along the optical axis **190**; the surfaces 4, 13 and 14 represent the thickness of the reflective polarizer **142** along the optical axis **190**; the surfaces 5, 12 and 15 represent the thickness of the first phase retarder **143** along the optical axis **190**; the surfaces 6 and 16 represent the thickness of the second lens **120** along the optical axis **190**; the surface 11 represents a gap between the image source-side surface **122** of the second lens **120** and the visual-side surface **121** of the second lens **120** along the optical axis **190**, this gap corresponds to the thickness of the second lens **120** along the optical axis **190**; the surfaces 7, 10 and 17 represent a gap between the image source-side surface **122** of the second lens **120** and the visual-side surface **131** of the third lens **130** along the optical axis **190**; the surfaces 8 and 18 represent the thickness of the third lens **130** along the optical axis **190**; the surface 9 represents a gap between the image source-side surface **132** of the third lens **130** and the visual-side surface **131** of the third lens **130** along the optical axis **190**, this gap corresponds to the thickness of the third lens **130** along the optical axis **190**; the surfaces 19 represents a gap between the image source-side surface **132** of the third lens **130** and the second phase retarder **161** along the optical axis **190**; the surface 20 represents the thickness of the second phase retarder **161** along the optical axis **190**; the surface 21 represents the thickness of the second absorptive polarizer **162** along the optical axis **190**; and the surface 22 is the image source plane **181**. The gaps and thicknesses having a positive sign in Table 1 denote the transmission direction of light is toward the stop **100**, and the gaps and thicknesses having a negative sign in Table 1 denote the transmission direction of light is toward the image source plane **181**.

In table 2, k represents the conic constant of the equation of aspheric surface profiles, and A2, A4, A6, A8, A10, A12, A14, A16, A18, and A20 represent the high-order aspheric coefficients.

The respective tables presented below for respective one of other embodiments are based on the schematic view of this embodiment, and the definitions of parameters in the tables are the same as those in Tables 1-4 of the first embodiment. However, the definition of each surface number in Table 1 varies with the number of the lenses and the position of the optical elements, and the relevant description of the embodiments may be referred to the definition mode of each surface number in Table 1. Therefore, an explanation in this regard will not be provided again.

Second Embodiment

Referring to FIG. 2, an optical lens assembly in accordance with a second embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **290**: a stop **200**, a first lens **210**, a first optical element group **240**, a second lens **220**, a third lens **230**, a partial-reflective-partial-transmissive element **250**, a second optical element group **260** and an image source plane **281**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop **200** may be located in a position where the user's eyes view an image.

The first lens **210** with positive refractive power includes a visual-side surface **211** and an image source-side surface **212**, the visual-side surface **211** of the first lens **210** is convex in a paraxial region thereof, the image source-side surface **212** of the first lens **210** is concave in a paraxial

region thereof, the visual-side surface **211** and the image source-side surface **212** of the first lens **210** are aspheric, and the first lens **210** is made of plastic.

The second lens **220** with negative refractive power includes a visual-side surface **221** and an image source-side surface **222**, the visual-side surface **221** of the second lens **220** is flat in a paraxial region thereof, the image source-side surface **222** of the second lens **220** is concave in a paraxial region thereof, the image source-side surface **222** of the second lens **220** is spherical, and the second lens **220** is made of plastic.

The third lens **230** with positive refractive power includes a visual-side surface **231** and an image source-side surface **232**, the visual-side surface **231** of the third lens **230** is convex in a paraxial region thereof, the image source-side surface **232** of the third lens **230** is convex in a paraxial region thereof, the visual-side surface **231** of the third lens **230** is spherical, the image source-side surface **232** of the third lens **230** is aspheric, and the third lens **230** is made of plastic. The third lens **230** and the second lens **220** together form a cemented doublet lens.

The first optical element group **240** is disposed on the visual-side surface **221** of the second lens **220** and includes, in order from the visual side to the image source side along the optical axis **290**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

The partial-reflective-partial-transmissive element **250** is disposed on the image source-side surface **232** of the third lens **230** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **250** for different wavelengths.

The second optical element group **260** is disposed on the image source plane **281** and includes, in order from the visual side to the image source side along the optical axis **290**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

The optical lens assembly works in cooperation with an image source **283** disposed on the image source plane **281**. In the present embodiment, the type of the image source **283** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

Please refer to Tables 5-8, Table 5 shows the detailed optical data of the elements of the optical lens assembly of the second embodiment, Table 6 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the second embodiment, Table 7 shows the remaining parameters of the optical lens assembly of the second embodiment and the values thereof, and the values of the parameters in Tables 5 and 7 meet the conditional formulas of Table 8. In the second embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 5 can be referred to the relevant description of Table 1 and will not be explained again.

TABLE 5

Embodiment 2						
f = 19.82 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.04°						
Surface		Radius of curvature	Thickness/gap	Refractive index (nd)	Abbe number (vd)	Refraction/reflection
0	Stop	Infinity	15.000	—	—	—
1	First lens	24.606	6.704	1.544	55.9	Refraction
2		126.414	4.096	—	—	Refraction
3	First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
4	Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction
5	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
6	Second lens	Infinity	2.000	1.645	23.400	Refraction
7	Third lens	62.105	4.662	1.544	55.9	Refraction
8	Partial-reflective-partial-transmissive element	-71.153	-4.662	—	—	Reflection
9	Second lens	62.105	-2.000	1.645	23.400	Refraction
10	First phase retarder	Infinity	-0.100	1.533	56.0	Refraction
11	Reflective polarizer	Infinity	-0.100	1.533	56.0	Refraction
12	Reflective polarizer	Infinity	0.100	—	—	Reflection
13	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
14	Second lens	Infinity	2.000	1.645	23.400	Refraction
15	Third lens	62.105	4.662	1.544	55.9	Refraction
16	Partial-reflective-partial-transmissive element	-71.153	6.038	—	—	Refraction
17	Second phase retarder	Infinity	0.100	1.533	56.0	Refraction
18	Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
19	Image source plane	Infinity	—	—	—	—

The reference wavelength is 550 nm.

TABLE 6

Embodiment 2						
Aspheric Coefficients						
Surface	1	2	6, 10, 14	7, 9, 15	7, 9, 15	8, 16
K:	-9.7435E+00	9.0000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	6.4634E-05	-3.0925E-05	0.0000E+00	0.0000E+00	0.0000E+00	5.0297E-06
A6:	-6.9913E-07	-3.6771E-07	0.0000E+00	0.0000E+00	0.0000E+00	1.2349E-08
A8:	4.9207E-09	1.9661E-09	0.0000E+00	0.0000E+00	0.0000E+00	1.5945E-10
A10:	-1.8438E-11	-9.7058E-12	0.0000E+00	0.0000E+00	0.0000E+00	-1.0803E-12
A12:	-1.5556E-13	-6.4398E-14	0.0000E+00	0.0000E+00	0.0000E+00	2.0297E-15
A14:	5.1182E-16	6.8206E-17	0.0000E+00	0.0000E+00	0.0000E+00	2.9897E-17
A16:	1.8205E-18	2.2308E-18	0.0000E+00	0.0000E+00	0.0000E+00	-1.3317E-19
A18:	5.3367E-20	5.4754E-21	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	-4.1168E-22	-7.8393E-23	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 7

Embodiment 2					
f1 [mm]	54.56	CA1 [mm]	11.07	TDP1 [mm]	1.95
f2 [mm]	-95.74	CA2 [mm]	12.07	TDP2 [mm]	0.97
f3 [mm]	61.35	—	—	—	—

TABLE 8

Embodiment 2			
f1/f	2.75	f3/f	3.10
R1/R2	0.19	R6/R5	-1.15
R6/R2	-0.56	R6/CT3	-15.26
CT3/CT2	2.33	(R1*R2)/f1	50.70
(CT1 + CT2)/CT3	1.87	CA1/TDP1	5.68
R1/CT1	3.67	CA2/TDP2	12.45
f3/R1	2.49	(CA2/TL)*R1	12.37

TABLE 8-continued

Embodiment 2			
f1/f3	0.89	(R2/CT1)(f1/f)	51.91
f2/f	-4.83	—	—

Third Embodiment

Referring to FIG. 3, an optical lens assembly in accordance with a third embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis 390: a stop 300, a first lens 310, a first optical element group 340, a second lens 320, a third lens 330, a partial-reflective-partial-transmissive element 350, a second optical element group 360 and an image source plane 381. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop 300 may be located in a position where the user's eyes view an image.

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The first lens **310** with positive refractive power includes a visual-side surface **311** and an image source-side surface **312**, the visual-side surface **311** of the first lens **310** is convex in a paraxial region thereof, the image source-side surface **312** of the first lens **310** is concave in a paraxial region thereof, the visual-side surface **311** and the image source-side surface **312** of the first lens **310** are aspheric, and the first lens **310** is made of plastic.

The second lens **320** with positive refractive power includes a visual-side surface **321** and an image source-side surface **322**, the visual-side surface **321** of the second lens **320** is flat in a paraxial region thereof, the image source-side surface **322** of the second lens **320** is convex in a paraxial region thereof, the image source-side surface **322** of the second lens **320** is spherical, and the second lens **320** is made of plastic.

The third lens **330** with negative refractive power includes a visual-side surface **331** and an image source-side surface **332**, the visual-side surface **331** of the third lens **330** is concave in a paraxial region thereof, the image source-side surface **332** of the third lens **330** is convex in a paraxial region thereof, the visual-side surface **331** of the third lens **330** is spherical, the image source-side surface **332** of the third lens **330** is aspheric, and the third lens **330** is made of plastic. The third lens **330** and the second lens **320** together form a cemented doublet lens.

The first optical element group **340** is disposed on the visual-side surface **321** of the second lens **320** and includes, in order from the visual side to the image source side along the optical axis **390**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

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The partial-reflective-partial-transmissive element **350** is disposed on the image source-side surface **332** of the third lens **330** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **350** for different wavelengths.

The second optical element group **360** is disposed on the image source plane **381** and includes, in order from the visual side to the image source side along the optical axis **390**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

The optical lens assembly works in cooperation with an image source **383** disposed on the image source plane **381**. In the present embodiment, the type of the image source **383** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

Please refer to Tables 9-12, Table 9 shows the detailed optical data of the elements of the optical lens assembly of the third embodiment, Table 10 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the third embodiment, Table 11 shows the remaining parameters of the optical lens assembly of the third embodiment and the values thereof, and the values of the parameters in Tables 9 and 11 meet the conditional formulas of Table 12. In the third embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 9 can be referred to the relevant description of Table 1 and will not be explained again.

TABLE 9

Embodiment 3						
f = 19.78 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.05°						
Surface		Radius of curvature	Thickness/gap	Refractive index (nd)	Abbe number (vd)	Refraction/reflection
0	Stop	Infinity	15.000	—	—	—
1	First lens	24.533	5.875	1.544	55.9	Refraction
2		126.787	4.805	—	—	Refraction
3	First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
4	Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction
5	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
6	Second lens	Infinity	4.413	1.544	55.9	Refraction
7	Third lens	-36.214	1.891	1.645	23.400	Refraction
8	Partial-reflective-partial-transmissive element	-72.358	-1.891	—	—	Reflection
9	Second lens	-36.214	-4.413	1.544	55.900	Refraction
10	First phase retarder	Infinity	-0.100	1.533	56.0	Refraction
11	Reflective polarizer	Infinity	-0.100	1.533	56.0	Refraction
12	Reflective polarizer	Infinity	0.100	—	—	Reflection
13	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
14	Second lens	Infinity	4.413	1.544	55.900	Refraction
15	Third lens	-36.214	1.891	1.544	55.9	Refraction
16	Partial-reflective-partial-transmissive element	-72.358	6.516	—	—	Refraction
17	Second phase retarder	Infinity	0.100	1.533	56.0	Refraction
18	Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
19	Image source plane	Infinity	—	—	—	—

The reference wavelength is 550 nm.

TABLE 10

Embodiment 3 Aspheric Coefficients						
Surface	1	2	6, 10, 14	7, 9, 15	7, 9, 15	8, 16
K:	-1.1894E+01	9.0000E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	6.6598E-05	-5.0930E-05	0.0000E+00	0.0000E+00	0.0000E+00	5.2297E-06
A6:	-7.5449E-07	-1.6412E-07	0.0000E+00	0.0000E+00	0.0000E+00	1.7442E-08
A8:	4.3204E-09	-5.4893E-10	0.0000E+00	0.0000E+00	0.0000E+00	-8.2624E-12
A10:	-2.0618E-11	-2.6110E-12	0.0000E+00	0.0000E+00	0.0000E+00	2.7055E-13
A12:	-1.0590E-13	-8.4544E-15	0.0000E+00	0.0000E+00	0.0000E+00	-1.0573E-15
A14:	1.0188E-15	7.4121E-18	0.0000E+00	0.0000E+00	0.0000E+00	9.4549E-19
A16:	2.0165E-19	7.1377E-19	0.0000E+00	0.0000E+00	0.0000E+00	-1.2703E-20
A18:	7.1422E-22	6.5305E-23	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	-2.0311E-22	-5.0571E-23	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 11

Embodiment 3					
f1 [mm]	54.48	CA1 [mm]	10.96	TDP1 [mm]	1.73
f2 [mm]	66.17	CA2 [mm]	11.86	TDP2 [mm]	1.16
f3 [mm]	-114.11	—	—	—	—

TABLE 12

Embodiment 3			
f1/f	2.75	f3/f	-5.77
R1/R2	0.19	R6/R5	2.00
R6/R2	-0.57	R6/CT3	-38.27
CT3/CT2	0.43	(R1*R2)/f1	-27.26
(CT1 + CT2)/CT3	5.44	CA1/TDP1	6.33
R1/CT1	4.18	CA2/TDP2	10.21
f3/R1	-4.65	(CA2/TL)*R1	12.13
f1/f3	-0.48	(R2/CT1)/(f1/f)	59.43
f2/f	3.35	—	—

Fourth Embodiment

Referring to FIG. 4, an optical lens assembly in accordance with a fourth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis 490: a stop 400, a first lens 410, a first optical element group 440, a second lens 420, a third lens 430, a partial-reflective-partial-transmissive element 450, a second optical element group 460 and an image source plane 481. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop 400 may be located in a position where the user's eyes view an image.

The first lens 410 with positive refractive power includes a visual-side surface 411 and an image source-side surface 412, the visual-side surface 411 of the first lens 410 is convex in a paraxial region thereof, the image source-side surface 412 of the first lens 410 is concave in a paraxial region thereof, the visual-side surface 411 and the image source-side surface 412 of the first lens 410 are aspheric, and the first lens 410 is made of plastic.

The second lens 420 with positive refractive power includes a visual-side surface 421 and an image source-side surface 422, the visual-side surface 421 of the second lens 420 is flat in a paraxial region thereof, the image source-side surface 422 of the second lens 420 is convex in a paraxial region thereof, the image source-side surface 422 of the second lens 420 is aspheric, and the second lens 420 is made of plastic.

The third lens 430 with negative refractive power includes a visual-side surface 431 and an image source-side surface

432, the visual-side surface 431 of the third lens 430 is concave in a paraxial region thereof, the image source-side surface 432 of the third lens 430 is convex in a paraxial region thereof, the visual-side surface 431 and the image source-side surface 432 of the third lens 430 are aspheric, and the third lens 430 is made of plastic.

The first optical element group 440 is disposed on the visual-side surface 421 of the second lens 420 and includes, in order from the visual side to the image source side along the optical axis 490: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer 141, the reflective polarizer 142 and the first phase retarder 143 of the first embodiment and will not be explained again.

The partial-reflective-partial-transmissive element 450 is disposed on the image source-side surface 432 of the third lens 430 and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element 450 for different wavelengths.

The second optical element group 460 is disposed on the image source plane 481 and includes, in order from the visual side to the image source side along the optical axis 490: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder 161 and the second absorptive polarizer 162 of the first embodiment and will not be explained again.

The optical lens assembly works in cooperation with an image source 483 disposed on the image source plane 481. In the present embodiment, the type of the image source 483 is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

Please refer to Tables 13-16, Table 13 shows the detailed optical data of the elements of the optical lens assembly of the fourth embodiment, Table 14 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the fourth embodiment, Table 15 shows the remaining parameters of the optical lens assembly of the fourth embodiment and the values thereof, and the values of the parameters in Tables 13 and 15 meet the conditional formulas of Table 16. In the fourth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 13 can be referred to the relevant description of Table 1 and will not be explained again.

TABLE 13

Embodiment 4						
f = 19.56 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 52.00°						
Surface		Radius of curvature	Thickness/gap	Refractive index (nd)	Abbe number (vd)	Refraction/reflection
0	Stop	Infinity	15.000	—	—	—
1	First lens	21.291	6.306	1.544	55.9	Refraction
2		43.174	1.025	—	—	Refraction
3	First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
4	Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction
5	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
6	Second lens	Infinity	6.269	1.544	55.9	Refraction
7		-21.760	0.428	—	—	Refraction
8	third lens	-27.707	1.740	1.544	55.900	Refraction
9	Partial-reflective-partial-transmissive element	-84.726	-1.740	—	—	Reflection
10		-21.760	-0.428	—	—	Refraction
11	Second lens	-27.707	-6.269	1.544	55.900	Refraction
12	First phase retarder	Infinity	-0.100	1.533	56.0	Refraction
13	Reflective polarizer	Infinity	-0.100	1.533	56.0	Refraction
14	Reflective polarizer	Infinity	0.100	—	—	Reflection
15	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
16	Second lens	Infinity	6.269	1.544	55.900	Refraction
17		-21.760	0.428	—	—	Refraction
18	Third lens	-27.707	1.740	1.544	55.9	Refraction
19	Partial-reflective-partial-transmissive element	-84.726	4.294	—	—	Refraction
20	Second phase retarder	Infinity	0.100	1.533	56.0	Refraction
21	Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
22	Image source plane	Infinity	—	—	—	—

The reference wavelength is 550 nm.

TABLE 14

Embodiment 4						
Aspheric Coefficients						
Surface	1	2	6, 12, 16	7, 10, 17	8, 11, 18	9, 19
K:	-2.5932E-01	-2.0810E+00	0.0000E+00	-8.1323E-01	-4.6842E-01	8.2000E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	-1.4555E-05	-1.7511E-05	0.0000E+00	2.9577E-06	-5.0664E-05	-2.8280E-06
A6:	3.8026E-08	-1.1557E-07	0.0000E+00	-5.0314E-08	3.9196E-10	3.2122E-08
A8:	-8.3422E-10	-6.3251E-10	0.0000E+00	-2.6171E-10	6.2425E-10	1.5667E-10
A10:	-4.3414E-12	-2.6505E-12	0.0000E+00	7.8191E-13	3.2496E-13	3.0556E-13
A12:	1.1171E-14	-1.9553E-14	0.0000E+00	1.3185E-14	5.7519E-15	-4.0344E-16
A14:	1.0867E-16	-7.9811E-17	0.0000E+00	6.2014E-17	9.9697E-17	-7.0206E-18
A16:	-2.0298E-18	7.0121E-19	0.0000E+00	2.7812E-20	2.6410E-19	9.1281E-20
A18:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 15

Embodiment 4					
f1 [mm]	69.65	CA1 [mm]	11.32	TDP1 [mm]	2.62
f2 [mm]	39.76	CA2 [mm]	12.17	TDP2 [mm]	0.26
f3 [mm]	-76.05	—	—	—	—

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TABLE 16-continued

Embodiment 4			
f3/R1	-3.57	(CA2/TL)*R1	12.60
f1/f3	-0.92	(R2/CT1)(f1/f)	24.38
f2/f	2.03	—	—

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TABLE 16

Embodiment 4			
f1/f	3.56	f3/f	-3.89
R1/R2	0.49	R6/R5	3.06
R6/R2	-1.96	R6/CT3	-48.70
CT3/CT2	0.28	(R1*R2)/f1	-12.09
(CT1 + CT2)/CT3	7.23	CA1/TDP1	4.33
R1/CT1	3.38	CA2/TDP2	46.80

60

Fifth Embodiment

Referring to FIG. 5, an optical lens assembly in accordance with a fifth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **590**: a stop **500**, a first lens **510**, a second lens **520**, a first optical element group **540**, a third lens **530**, a partial-reflective-partial-transmissive element **550**, a second optical element group **560** and an image

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source plane **581**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop **500** may be located in a position where the user's eyes view an image.

The first lens **510** with positive refractive power includes a visual-side surface **511** and an image source-side surface **512**, the visual-side surface **511** of the first lens **510** is convex in a paraxial region thereof, the image source-side surface **512** of the first lens **510** is concave in a paraxial region thereof, the visual-side surface **511** and the image source-side surface **512** of the first lens **510** are aspheric, and the first lens **510** is made of plastic.

The second lens **520** with positive refractive power includes a visual-side surface **521** and an image source-side surface **522**, the visual-side surface **521** of the second lens **520** is convex in a paraxial region thereof, the image source-side surface **522** of the second lens **520** is concave in a paraxial region thereof, the visual-side surface **521** and the image source-side surface **522** of the second lens **520** are aspheric, and the second lens **520** is made of plastic.

The third lens **530** with positive refractive power includes a visual-side surface **531** and an image source-side surface **532**, the visual-side surface **531** of the third lens **530** is flat in a paraxial region thereof, the image source-side surface **532** of the third lens **530** is convex in a paraxial region thereof, the image source-side surface **532** of the third lens **530** is aspheric, and the third lens **530** is made of plastic.

The first optical element group **540** is disposed on the visual-side surface **531** of the third lens **530** and includes, in order from the visual side to the image source side along the optical axis **590**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

The partial-reflective-partial-transmissive element **550** is disposed on the image source-side surface **532** of the third lens **530** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **550** for different wavelengths.

The second optical element group **560** is disposed on the image source plane **581** and includes, in order from the visual side to the image source side along the optical axis **590**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

The optical lens assembly works in cooperation with an image source **583** disposed on the image source plane **581**. In the present embodiment, the type of the image source **583** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

Please refer to Tables 17-20, Table 17 shows the detailed optical data of the elements of the optical lens assembly of the fifth embodiment, Table 18 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the fifth embodiment, Table 19 shows the remaining parameters of the optical lens assembly of the fifth embodiment and the values thereof, and the values of the parameters in Tables 17 and 19 meet the conditional formulas of Table 20. In the fifth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 17 can be referred to the relevant description of Table 1 and will not be explained again.

TABLE 17

Embodiment 5						
f = 18.17 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 56.00°						
Surface		Radius of curvature	Thickness/gap	Refractive index (nd)	Abbe number (vd)	Refraction/reflection
0	Stop	Infinity	15.000	—	—	—
1	First lens	16.833	4.979	1.544	55.9	Refraction
2		17.373	1.204	—	—	Refraction
3	Second lens	24.044	4.664	1.544	55.9	Refraction
4		102.772	2.000	—	—	Refraction
5	First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
6	Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction
7	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
8	Third lens	Infinity	5.784	1.544	55.900	Refraction
9	Partial-reflective-partial-transmissive element	-66.430	-5.784	—	—	Reflection
10	First phase retarder	Infinity	-0.100	1.533	56.0	Refraction
11	Reflective polarizer	Infinity	-0.100	1.533	56.0	Refraction
12	Reflective polarizer	Infinity	0.100	—	—	Reflection
13	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
14	Third lens	Infinity	5.784	1.544	55.9	Refraction
15	Partial-reflective-partial-transmissive element	-66.430	4.870	—	—	Refraction
16	Second phase retarder	Infinity	0.100	1.533	56.0	Refraction
17	Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
18	Image source plane	Infinity	—	—	—	—

The reference wavelength is 550 nm.

TABLE 18

Embodiment 5 Aspheric Coefficients						
Surface	1	2	3	4	8, 10, 14	9, 15
K:	-2.4483E-01	3.6440E-02	-5.3088E-02	-6.3630E+01	0.0000E+00	-1.2456E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	-1.3939E-05	-1.3675E-05	-3.8894E-06	-8.0253E-06	0.0000E+00	5.9760E-06
A6:	2.3663E-09	-7.1810E-08	-8.5553E-09	-2.6639E-08	0.0000E+00	1.2082E-08
A8:	-3.1998E-10	-3.7038E-10	-2.5552E-11	4.1260E-11	0.0000E+00	-1.1398E-11
A10:	-1.0722E-12	-1.2803E-12	4.4043E-13	4.1232E-13	0.0000E+00	8.5732E-13
A12:	5.0972E-15	-2.4952E-15	-1.4351E-16	9.0535E-16	0.0000E+00	-4.6428E-16
A14:	4.5081E-18	-8.4687E-18	-6.0146E-18	4.0754E-19	0.0000E+00	-4.7947E-17
A16:	-8.9487E-20	1.2044E-20	6.2660E-20	-8.9767E-23	0.0000E+00	2.7543E-19
A18:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 19

Embodiment 5					
f1 [mm]	232.24	CA1 [mm]	13.61	TDP1 [mm]	5.47
f2 [mm]	56.18	CA2 [mm]	13.66	TDP2 [mm]	4.88
f3 [mm]	121.38	—	—	—	—

TABLE 20

Embodiment 5			
f1/f	12.78	f3/f	6.68
R1/R2	0.97	R6/R5	0.00
R6/R2	-3.82	R6/CT3	-11.49
CT3/CT2	1.24	(R1*R2)/f1	2.41
(CT1 + CT2)/CT3	1.67	CA1/TDP1	2.49
R1/CT1	3.38	CA2/TDP2	2.80
f3/R1	7.21	(CA2/TL)*R1	9.58
f1/f3	1.91	(R2/CT1)(f1/f)	44.59
f2/f	3.09	—	—

Sixth Embodiment

Referring to FIG. 6, an optical lens assembly in accordance with a sixth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis 690: a stop 600, a first lens 610, a second lens 620, a first optical element group 640, a third lens 630, a partial-reflective-partial-transmissive element 650, a second optical element group 660 and an image source plane 681. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop 600 may be located in a position where the user's eyes view an image.

The first lens 610 with positive refractive power includes a visual-side surface 611 and an image source-side surface 612, the visual-side surface 611 of the first lens 610 is convex in a paraxial region thereof, the image source-side surface 612 of the first lens 610 is convex in a paraxial region thereof, the visual-side surface 611 and the image source-side surface 612 of the first lens 610 are aspheric, and the first lens 610 is made of plastic.

The second lens 620 with negative refractive power includes a visual-side surface 621 and an image source-side surface 622, the visual-side surface 621 of the second lens 620 is concave in a paraxial region thereof, the image source-side surface 622 of the second lens 620 is convex in a paraxial region thereof, the visual-side surface 621 and the image source-side surface 622 of the second lens 620 are aspheric, and the second lens 620 is made of plastic.

The third lens 630 with positive refractive power includes a visual-side surface 631 and an image source-side surface 632, the visual-side surface 631 of the third lens 630 is flat in a paraxial region thereof, the image source-side surface 632 of the third lens 630 is convex in a paraxial region thereof, the image source-side surface 632 of the third lens 630 is spherical, and the third lens 630 is made of plastic.

The first optical element group 640 is disposed on the visual-side surface 631 of the third lens 630 and includes, in order from the visual side to the image source side along the optical axis 690: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer 141, the reflective polarizer 142 and the first phase retarder 143 of the first embodiment and will not be explained again.

The partial-reflective-partial-transmissive element 650 is disposed on the image source-side surface 632 of the third lens 630 and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element 650 for different wavelengths.

The second optical element group 660 is disposed on the image source plane 681 and includes, in order from the visual side to the image source side along the optical axis 690: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder 161 and the second absorptive polarizer 162 of the first embodiment and will not be explained again.

The optical lens assembly works in cooperation with an image source 683 disposed on the image source plane 681. In the present embodiment, the type of the image source 683 is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

Please refer to Tables 21-24, Table 21 shows the detailed optical data of the elements of the optical lens assembly of the sixth embodiment, Table 22 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the sixth embodiment, Table 23 shows the remaining parameters of the optical lens assembly of the sixth embodiment and the values thereof, and the values of the parameters in Tables 21 and 23 meet the conditional formulas of Table 24. In the sixth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 21 can be referred to the relevant description of Table 1 and will not be explained again.

TABLE 21

Embodiment 6						
f = 18.14 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 60.52°						
Surface		Radius of curvature	Thickness/gap	Refractive Index (nd)	Abbe number (vd)	Refraction/reflection
0	Stop	Infinity	15.000	—	—	—
1	First lens	67.587	8.308	1.544	55.9	Refraction
2		-22.871	0.964	—	—	Refraction
3	Second lens	-13.410	2.710	1.544	55.9	Refraction
4		-19.512	0.100	—	—	Refraction
5	First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
6	Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction
7	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
8	Third lens	Infinity	10.418	1.544	55.900	Refraction
9	Partial-reflective-partial-transmissive element	-69.351	-10.418	—	—	Reflection
10	First phase retarder	Infinity	-0.100	1.533	56.0	Refraction
11	Reflective polarizer	Infinity	-0.100	1.533	56.0	Refraction
12	Reflective polarizer	Infinity	0.100	—	—	Reflection
13	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
14	Third lens	Infinity	10.418	1.544	55.9	Refraction
15	Partial-reflective-partial-transmissive element	-69.351	1.000	—	—	Refraction
16	Second phase retarder	Infinity	0.100	1.533	56.0	Refraction
17	Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
18	Image source plane	Infinity	—	—	—	—

The reference wavelength is 550 nm.

TABLE 22

Embodiment 6						
Aspheric Coefficients						
Surface	1	2	3	4	8, 10, 14	9, 15
K:	0.0000E+00	0.0000E+00	-2.0627E-01	0.0000E+00	0.0000E+00	0.0000E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	-1.1715E-05	-4.0874E-05	5.8026E-05	6.4153E-05	0.0000E+00	0.0000E+00
A6:	3.7663E-11	-8.0525E-08	1.6400E-07	1.0691E-07	0.0000E+00	0.0000E+00
A8:	-7.2412E-10	7.0159E-11	2.0777E-10	-1.9260E-10	0.0000E+00	0.0000E+00
A10:	-7.5971E-12	2.4966E-12	6.1292E-12	-8.0932E-13	0.0000E+00	0.0000E+00
A12:	-1.4095E-14	1.8405E-14	2.5661E-15	7.6759E-16	0.0000E+00	0.0000E+00
A14:	4.4519E-16	-2.5695E-16	-1.3272E-16	1.2401E-17	0.0000E+00	0.0000E+00
A16:	-3.9216E-18	0.0000E+00	5.8205E-19	-1.4243E-20	0.0000E+00	0.0000E+00
A18:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 23

Embodiment 6					
f1 [mm]	32.27	CA1 [mm]	11.54	TDP1 [mm]	0.10
f2 [mm]	-80.09	CA2 [mm]	12.96	TDP2 [mm]	5.73
f3 [mm]	126.72	—	—	—	—

TABLE 24-continued

Embodiment 6			
f1/f3	0.25	(R2/CT1)(f1/f)	-4.90
f2/f	-4.41	—	—

Seventh Embodiment

TABLE 24

Embodiment 6			
f1/f	1.78	f3/f	6.99
R1/R2	-2.96	R6/R5	0.00
R6/R2	3.03	R6/CT3	-6.66
CT3/CT2	3.84	(R1*R2)/f1	-12.20
(CT1 + CT2)/CT3	1.06	CA1/TDP1	114.31
R1/CT1	8.14	CA2/TDP2	2.26
f3/R1	1.87	(CA2/TL)*R1	36.50

Referring to FIG. 7, an optical lens assembly in accordance with a seventh embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **790**: a stop **700**, a first lens **710**, a second lens **720**, a first optical element group **740**, a third lens **730**, a partial-reflective-partial-transmissive element **750**, a second optical element group **760** and an image source plane **781**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop **700** may be located in a position where the user's eyes view an image.

The first lens **710** with positive refractive power includes a visual-side surface **711** and an image source-side surface **712**, the visual-side surface **711** of the first lens **710** is concave in a paraxial region thereof, the image source-side surface **712** of the first lens **710** is convex in a paraxial region thereof, the visual-side surface **711** and the image source-side surface **712** of the first lens **710** are aspheric, and the first lens **710** is made of plastic.

The second lens **720** with negative refractive power includes a visual-side surface **721** and an image source-side surface **722**, the visual-side surface **721** of the second lens **720** is concave in a paraxial region thereof, the image source-side surface **722** of the second lens **720** is convex in a paraxial region thereof, the visual-side surface **721** and the image source-side surface **722** of the second lens **720** are aspheric, and the second lens **720** is made of plastic. The second lens **720** and the first lens **710** together form a cemented doublet lens.

The third lens **730** with positive refractive power includes a visual-side surface **731** and an image source-side surface **732**, the visual-side surface **731** of the third lens **730** is flat in a paraxial region thereof, the image source-side surface **732** of the third lens **730** is convex in a paraxial region thereof, the image source-side surface **732** of the third lens **730** is spherical, and the third lens **730** is made of plastic.

The first optical element group **740** is disposed on the visual-side surface **731** of the third lens **730** and includes, in order from the visual side to the image source side along the optical axis **790**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

The partial-reflective-partial-transmissive element **750** is disposed on the image source-side surface **732** of the third lens **730** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **750** for different wavelengths.

The second optical element group **760** is disposed on the image source plane **781** and includes, in order from the visual side to the image source side along the optical axis **790**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

The optical lens assembly works in cooperation with an image source **783** disposed on the image source plane **781**. In the present embodiment, the type of the image source **783** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

Please refer to Tables 25-28, Table 25 shows the detailed optical data of the elements of the optical lens assembly of the seventh embodiment, Table 26 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the seventh embodiment, Table 27 shows the remaining parameters of the optical lens assembly of the seventh embodiment and the values thereof, and the values of the parameters in Tables 25 and 27 meet the conditional formulas of Table 28. In the seventh embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 25 can be referred to the relevant description of Table 1 and will not be explained again.

TABLE 25

Embodiment 7						
f = 19.57 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 56.00°						
Surface	Radius of curvature	Thickness/gap	Refractive index (nd)	Abbe number (vd)	Refraction/reflection	
0 Stop	Infinity	15.000	—	—	—	
1 First lens	−87.167	7.762	1.544	55.9	Refraction	
2	−16.051	0.000	—	—	Refraction	
3 Second lens	−16.051	2.000	1.645	23.4	Refraction	
4	−23.678	0.100	—	—	Refraction	
5 First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction	
6 Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction	
7 First phase retarder	Infinity	0.100	1.533	56.0	Refraction	
8 Third lens	Infinity	12.638	1.544	55.900	Refraction	
9 Partial-reflective-partial-transmissive element	−81.316	−12.638	—	—	Reflection	
10 First phase retarder	Infinity	−0.100	1.533	56.0	Refraction	
11 Reflective polarizer	Infinity	−0.100	1.533	56.0	Refraction	
12 Reflective polarizer	Infinity	0.100	—	—	Reflection	
13 First phase retarder	Infinity	0.100	1.533	56.0	Refraction	
14 Third lens	Infinity	12.638	1.544	55.9	Refraction	
15 Partial-reflective-partial-transmissive element	−81.316	1.000	—	—	Refraction	
16 Second phase retarder	Infinity	0.100	1.533	56.0	Refraction	
17 Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction	
18 Image source plane	Infinity	—	—	—	—	

The reference wavelength is 550 nm.

TABLE 26

Embodiment 7 Aspheric Coefficients						
Surface	1	2	3	4	8, 10, 14	9, 15
K:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	-2.3306E-05	-1.9255E-04	-1.9255E-04	-3.5158E-05	0.0000E+00	0.0000E+00
A6:	-2.6725E-07	1.4129E-06	1.4129E-06	1.2565E-07	0.0000E+00	0.0000E+00
A8:	1.9968E-09	-3.9152E-09	-3.9152E-09	-4.6121E-10	0.0000E+00	0.0000E+00
A10:	-2.3610E-11	-2.9871E-11	-2.9871E-11	-4.1219E-12	0.0000E+00	0.0000E+00
A12:	-4.9339E-14	3.8751E-14	3.8751E-14	8.8672E-15	0.0000E+00	0.0000E+00
A14:	1.9228E-15	8.2701E-16	8.2701E-16	1.0410E-16	0.0000E+00	0.0000E+00
A16:	-1.1411E-17	0.0000E+00	0.0000E+00	-4.7844E-19	0.0000E+00	0.0000E+00
A18:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 27

Embodiment 7					
f1 [mm]	34.61	CA1 [mm]	10.11	TDP1 [mm]	1.13
f2 [mm]	-85.66	CA2 [mm]	11.37	TDP2 [mm]	6.37
f3 [mm]	148.59	—	—	—	—

TABLE 28

Embodiment 7			
f1/f	1.77	f3/f	7.59
R1/R2	5.43	R6/R5	0.00
R6/R2	5.07	R6/CT3	-6.43
CT3/CT2	6.32	(R1*R2)/f1	9.42
(CT1 + CT2)/CT3	0.77	CA1/TDP1	8.94
R1/CT1	-11.23	CA2/TDP2	1.78
f3/R1	-1.70	(CA2/TL)*R1	-41.28
f1/f3	0.23	(R2/CT1)(f1/f)	-3.66
f2/f	-4.38	—	—

Eighth Embodiment

Referring to FIG. 8, an optical lens assembly in accordance with an eighth embodiment of the present invention includes, in order from a visual side to an image source side along an optical axis **890**: a stop **800**, a first lens **810**, a second lens **820**, a first optical element group **840**, a third lens **830**, a partial-reflective-partial-transmissive element **850**, a second optical element group **860** and an image source plane **881**. The optical lens assembly has a total of three lenses with refractive power, but not is limited thereto.

The stop **800** may be located in a position where the user's eyes view an image.

The first lens **810** with positive refractive power includes a visual-side surface **811** and an image source-side surface **812**, the visual-side surface **811** of the first lens **810** is convex in a paraxial region thereof, the image source-side surface **812** of the first lens **810** is concave in a paraxial region thereof, the visual-side surface **811** and the image source-side surface **812** of the first lens **810** are aspheric, and the first lens **810** is made of plastic.

The second lens **820** with positive refractive power includes a visual-side surface **821** and an image source-side surface **822**, the visual-side surface **821** of the second lens **820** is convex in a paraxial region thereof, the image source-side surface **822** of the second lens **820** is flat in a paraxial region thereof, the visual-side surface **821** of the second lens **820** is aspheric, and the second lens **820** is made of plastic.

The third lens **830** with positive refractive power includes a visual-side surface **831** and an image source-side surface **832**, the visual-side surface **831** of the third lens **830** is convex in a paraxial region thereof, the image source-side surface **832** of the third lens **830** is convex in a paraxial region thereof, the visual-side surface **831** and the image source-side surface **832** of the third lens **830** are aspheric, and the third lens **830** is made of plastic.

The first optical element group **840** is disposed on the image source-side surface **822** of the second lens **820** and includes, in order from the visual side to the image source side along the optical axis **890**: a first absorptive polarizer, a reflective polarizer and a first phase retarder. The configuration of the first absorptive polarizer, the reflective polarizer and the first phase retarder is the same as that of the first absorptive polarizer **141**, the reflective polarizer **142** and the first phase retarder **143** of the first embodiment and will not be explained again.

The partial-reflective-partial-transmissive element **850** is disposed on the image source-side surface **832** of the third lens **830** and has an average reflectance of at least 30%, preferably 50%, in the wavelength range of visible light. The average reflectance here is an average value of different reflectance of the partial-reflective-partial-transmissive element **850** for different wavelengths.

The second optical element group **860** is disposed on the image source plane **881** and includes, in order from the visual side to the image source side along the optical axis **890**: a second phase retarder and a second absorptive polarizer. The configuration of the second phase retarder and the second absorptive polarizer is the same as that of the second phase retarder **161** and the second absorptive polarizer **162** of the first embodiment and will not be explained again.

The optical lens assembly works in cooperation with an image source **883** disposed on the image source plane **881**. In the present embodiment, the type of the image source **883** is, for example, but not limited to, an OLED display, a LED display, a liquid crystal display, or other displays.

Please refer to Tables 29-32, Table 29 shows the detailed optical data of the elements of the optical lens assembly of the eighth embodiment, Table 30 shows the data of the aspherical surfaces of the lenses of the optical lens assembly of the eighth embodiment, Table 31 shows the remaining parameters of the optical lens assembly of the eighth embodiment and the values thereof, and the values of the parameters in Tables 29 and 31 meet the conditional formulas of Table 32. In the eighth embodiment, the equation of the aspheric surface profiles of the aforementioned lenses is the same as the equation of the aspheric surface profiles of the aforementioned lenses in the first embodiment. The definitions of the surfaces in Table 29 can be referred to the relevant description of Table 1 and will not be explained again.

TABLE 29

Embodiment 8						
f = 18.19 mm, EPD(Entrance pupil diameter) = 10.00 mm, FOV (Field of view) = 57.71°						
Surface		Radius of curvature	Thickness/ gap	Refractive index (nd)	Abbe number (vd)	Refraction/ reflection
0	Stop	Infinity	15.000	—	—	—
1	First lens	34.173	6.274	1.544	55.9	Refraction
2		42.937	4.857	—	—	Refraction
3	Second lens	110.714	2.135	1.544	55.9	Refraction
4	First absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
5	Reflective polarizer	Infinity	0.100	1.533	56.0	Refraction
6	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
7		Infinity	1.890	—	—	Refraction
8	Third lens	129.300	7.063	1.544	55.900	Refraction
9	Partial-reflective-partial-transmissive element	-68.753	-7.063	—	—	Reflection
10		129.300	-1.890	—	—	Refraction
11	First phase retarder	Infinity	-0.100	1.533	56.0	Refraction
12	Reflective polarizer	Infinity	-0.100	1.533	56.0	Refraction
13	Reflective polarizer	Infinity	0.100	—	—	Reflection
14	First phase retarder	Infinity	0.100	1.533	56.0	Refraction
15		Infinity	1.890	—	—	Refraction
16	Third lens	129.300	7.063	1.544	55.9	Refraction
17	Partial-reflective-partial-transmissive element	-68.753	2.280	—	—	Refraction
18	Second phase retarder	Infinity	0.100	1.533	56.0	Refraction
19	Second absorptive polarizer	Infinity	0.100	1.533	56.0	Refraction
20	Image source plane	Infinity	—	—	—	—

The reference wavelength is 550 nm.

TABLE 30

Embodiment 8						
Aspheric Coefficients						
Surface	1	2	3	4	8, 10, 16	9, 17
K:	-5.3525E+00	-7.5221E+00	-1.7631E+01	0.0000E+00	-3.0407E+01	-1.1050E+00
A2:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A4:	-6.9119E-06	-7.6849E-06	-7.9976E-08	0.0000E+00	-1.4962E-06	3.5987E-07
A6:	-4.9302E-08	-2.7343E-08	1.1354E-08	0.0000E+00	-8.2575E-10	1.0984E-09
A8:	-8.7928E-11	-1.0237E-10	9.2473E-12	0.0000E+00	-7.7597E-14	2.3102E-12
A10:	-6.0233E-13	-2.3540E-13	-1.2075E-14	0.0000E+00	4.5237E-15	1.7381E-15
A12:	-1.6016E-15	-2.0726E-17	-5.7442E-17	0.0000E+00	-1.8839E-18	2.8049E-18
A14:	1.8562E-17	1.6676E-18	-1.3546E-19	0.0000E+00	-4.6076E-20	1.3546E-20
A16:	-8.8664E-20	-4.9534E-21	3.1761E-22	0.0000E+00	3.2783E-22	2.7089E-23
A18:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
A20:	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00

TABLE 31

Embodiment 8					
f1 [mm]	244.11	CA1 [mm]	14.03	TDP1 [mm]	1.45
f2 [mm]	202.30	CA2 [mm]	16.37	TDP2 [mm]	0.70
f3 [mm]	83.06	—	—	—	—

TABLE 32

Embodiment 8			
f1/f	13.42	f3/f	4.57
R1/R2	0.80	R6/R5	0.00

TABLE 32-continued

Embodiment 8			
R6/R2	-1.60	R6/CT3	-9.73
CT3/CT2	3.31	(R1*R2)/f1	17.66
(CT1 + CT2)/CT3	1.19	CA1/TDP1	9.67
R1/CT1	5.45	CA2/TDP2	23.46
f3/R1	2.43	(CA2/TL)*R1	22.38
f1/f3	2.94	(R2/CT1)(f1/f)	91.85
f2/f	11.12	—	—

65 For the optical lens assembly in the present invention, the lenses can be made of plastic or glass. If the lens is made of plastic, it is conducive to reducing the manufacturing cost.

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If the lens is made of glass, it is conducive to enhancing the degree of freedom in the arrangement of refractive power of the optical lens assembly.

For the optical lens assembly in the present invention, the aspheric surface can have any profile shape other than the profile shape of a spherical surface, so more variables can be used in the design of aspheric surfaces (than spherical surfaces), which is conducive to reducing the aberration and the number of lenses, as well as the total length of the optical lens assembly.

For the optical lens assembly in the present invention, if the surface shape of a respective lens surface of a respective lens with refractive power is convex and the location of the convex portion of the respective lens surface of the respective lens is not defined, the convex portion is typically located in a paraxial region of the respective lens surface of the respective lens. If the surface shape of a respective lens surface of a respective lens is concave and the location of the concave portion of the respective lens surface of the respective lens is not defined, the concave portion is typically located in a paraxial region of the respective lens surface of the respective lens.

For the optical lens assembly in the present invention, the maximum effective radius of the lens surface is usually a radius of the effective optical region of the lens surface (i.e., a region which is not subjected to any surface treatment or extinction processing or is not provided with any shade).

The optical lens assembly of the present invention can be used in head-mounted electronic devices as required. FIG. 9 shows a head-mounted electronic device in accordance with an embodiment of the present invention. The head-mounted electronic device 9 is a head-mounted display device using, but not limited to, virtual reality (VR) technology, and includes a housing 910, an optical module 920, an image source 930 and a controller 940.

The optical module 920 corresponds to the left and right eyes of the user. The optical module 920 includes an optical lens assembly described in any one of the first to eighth embodiments.

The image source 930 can be an image source described in any one of the first to eighth embodiments. The image source 930 corresponds to the left and right eyes of the user, and the type of the image source 930 may be an OLED display, a LED display, a liquid crystal display, or other display, but is not limited thereto.

The controller 940 is electrically connected to the image source 930, so as to control the image source 930 to display an image, whereby the head-mounted electronic device 9 can project the image to the eyes of the user.

What is claimed is:

1. An optical lens assembly, comprising:
an optical element group; and

in order from a visual side to an image source side: a first lens with positive refractive power, a second lens with refractive power, a third lens with refractive power, and an image source-side surface of the third lens being convex in a paraxial region thereof, and a partial-reflective-partial-transmissive element;

wherein the optical element group is disposed between the first lens and the third lens, and includes, in order from the visual side to the image source side: an absorptive polarizer, a reflective polarizer and a phase retarder, a radius of curvature of an image source-side surface of the first lens is R2, a thickness of the first lens along an optical axis is CT1, a focal length of the first lens is f1,

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a focal length of the optical lens assembly is f, and the following condition is satisfied: $-5.88 < (R2/CT1) * (f1/f) < 110.22$.

2. The optical lens assembly as claimed in claim 1, wherein a focal length of the third lens is f3, the focal length of the optical lens assembly is f, and the following condition is satisfied: $-6.92 < f3/f < 23.80$.

3. The optical lens assembly as claimed in claim 1, wherein a focal length of the third lens is f3, a radius of curvature of a visual-side surface of the first lens is R1, and the following condition is satisfied: $-5.58 < f3/R1 < 27.18$.

4. The optical lens assembly as claimed in claim 1, wherein a maximum effective radius of a visual-side surface of the first lens is CA1, an absolute value of a displacement in parallel to the optical axis from an intersection between the visual-side surface of the first lens and the optical axis to the maximum effective radius position on the visual-side surface of the first lens is TDP1, and the following condition is satisfied: $1.76 < CA1/TDP1 < 137.17$.

5. The optical lens assembly as claimed in claim 1, wherein a maximum effective radius of the image source-side surface of the first lens is CA2, an absolute value of a displacement in parallel to the optical axis from an intersection between the image source-side surface of the first lens and the optical axis to the maximum effective radius position on the image source-side surface of the first lens is TDP2, and the following condition is satisfied: $1.43 < CA2/TDP2 < 56.16$.

6. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of the image source-side surface of the third lens is R6, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: $-58.44 < R6/CT3 < -5.15$.

7. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of a visual-side surface of the first lens is R1, the radius of curvature of the image source-side surface of the first lens is R2, the focal length of the first lens is f1, and the following condition is satisfied: $-32.71 \text{ mm} < (R1 * R2)/f1 < 60.84 \text{ mm}$.

8. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of a visual-side surface of the first lens is R1, the radius of curvature of the image source-side surface of the first lens is R2, and the following condition is satisfied: $-3.55 < R1/R2 < 6.52$.

9. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of the image source-side surface of the third lens is R6, the radius of curvature of the image source-side surface of the first lens is R2, and the following condition is satisfied: $-4.59 < R6/R2 < 6.08$.

10. The optical lens assembly as claimed in claim 1, wherein a thickness of the second lens along the optical axis is CT2, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: $0.22 < CT3/CT2 < 7.58$.

11. The optical lens assembly as claimed in claim 1, wherein the thickness of the first lens along the optical axis is CT1, a thickness of the second lens along the optical axis is CT2, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: $0.62 < (CT1 + CT2)/CT3 < 8.67$.

12. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of a visual-side surface of the first lens is R1, the thickness of the first lens along an optical axis is CT1, and the following condition is satisfied: $-13.48 < R1/CT1 < 9.76$.

13. The optical lens assembly as claimed in claim 1, wherein the focal length of the first lens is f1, the focal

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length of the optical lens assembly is f , and the following condition is satisfied: $1.42 < f_1/f < 16.1$.

14. The optical lens assembly as claimed in claim 1, wherein a maximum effective radius of the image source-side surface of the first lens is CA_2 , a distance from a visual-side surface of the first lens to an image source plane along the optical axis is TL , a radius of curvature of the visual-side surface of the first lens is R_1 , and the following condition is satisfied: $-49.53 \text{ mm} < (CA_2/TL) * R_1 < 43.8 \text{ mm}$.

15. A head-mounted electronic device, comprising:

a housing;

an optical lens assembly disposed in the housing;

an image source disposed on an image source plane of the optical lens assembly in the housing; and

a controller disposed in the housing and electrically connected to the image source;

wherein the optical lens assembly comprising:

an optical element group; and

in order from a visual side to an image source side: a first lens with positive refractive power, a second lens with refractive power, a third lens with refractive power, and an image source-side surface of the third lens being convex in a paraxial region thereof, and a partial-reflective-partial-transmissive element;

wherein the optical element group is disposed between the first lens and the third lens, and includes, in order from the visual side to the image source side: an absorptive polarizer, a reflective polarizer and a phase retarder, a radius of curvature of an image source-side surface of the first lens is R_2 , a thickness of the first lens along an optical axis is CT_1 , a focal length of the first lens is f_1 , a focal length of the optical lens assembly is f , and the following condition is satisfied: $-5.88 < (R_2/CT_1) * (f_1/f) < 110.22$.

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16. The head-mounted electronic device as claimed in claim 15, wherein a focal length of the third lens is f_3 , the focal length of the optical lens assembly is f , and the following condition is satisfied: $-6.92 < f_3/f < 23.80$.

17. The head-mounted electronic device as claimed in claim 15, wherein a maximum effective radius of a visual-side surface of the first lens is CA_1 , an absolute value of a displacement in parallel to the optical axis from an intersection between the visual-side surface of the first lens and the optical axis to the maximum effective radius position on the visual-side surface of the first lens is TDP_1 , and the following condition is satisfied: $1.76 < CA_1/TDP_1 < 137.17$.

18. The head-mounted electronic device as claimed in claim 15, wherein a radius of curvature of a visual-side surface of the first lens is R_1 , the radius of curvature of the image source-side surface of the first lens is R_2 , the focal length of the first lens is f_1 , and the following condition is satisfied: $-32.71 \text{ mm} < (R_1 * R_2)/f_1 < 60.84 \text{ mm}$.

19. The head-mounted electronic device as claimed in claim 15, wherein the thickness of the first lens along the optical axis is CT_1 , a thickness of the second lens along the optical axis is CT_2 , a thickness of the third lens along the optical axis is CT_3 , and the following condition is satisfied: $0.62 < (CT_1 + CT_2)/CT_3 < 8.67$.

20. The head-mounted electronic device as claimed in claim 15, wherein a maximum effective radius of the image source-side surface of the first lens is CA_2 , a distance from a visual-side surface of the first lens to the image source plane along the optical axis is TL , a radius of curvature of the visual-side surface of the first lens is R_1 , and the following condition is satisfied: $-49.53 \text{ mm} < (CA_2/TL) * R_1 < 43.8 \text{ mm}$.

* * * * *